

Learning Objectives

1. Understand the meaning of instantaneous and average power, use AC power notation, compute average power, and compute the power factor of a complex load. *Section 7.1.*
2. Use complex power notation; compute apparent, real, and reactive power for complex loads; and draw a power triangle. *Section 7.2.*
3. Compute the capacitance required to correct the power factor of a complex load. *Section 7.3.*
4. Analyze an ideal transformer; compute primary and secondary currents, voltages, and turns ratios; calculate reflected sources and impedances across ideal transformers; and understand maximum power transfer. *Section 7.4.*
5. Use three-phase AC power notation; and compute load currents and voltages for balanced wye and delta loads. *Section 7.5.*
6. Understand the basic principles of residential electrical wiring and of electrical safety. *Sections 7.6 and 7.7.*

7.1 INSTANTANEOUS AND AVERAGE POWER

When a linear electric circuit is excited by a sinusoidal source, all voltages and currents in the circuit are also sinusoids of the same frequency as the source. Figure 7.1 depicts the general form of a linear AC circuit. The most general expressions for the voltage and current delivered to an arbitrary load are as follows:

$$\begin{aligned} v(t) &= V \cos(\omega t + \theta_V) \\ i(t) &= I \cos(\omega t + \theta_I) \end{aligned} \quad (7.1)$$

where V and I are the peak amplitudes of the sinusoidal voltage and current, respectively, and θ_V and θ_I are their phase angles. Two such waveforms are plotted in Figure 7.2, with unit amplitude, angular frequency 150 rad/s, and phase angles $\theta_V = 0$ and $\theta_I = \pi/3$. Notice that the current *leads* the voltage; or equivalently, the voltage *lags* the current. Keep in mind that all phase angles are relative to some reference, which is usually chosen to be the phase angle of a source. The reference phase angle is freely chosen and therefore usually set to zero for simplicity. Also keep in mind that a phase angle represents a *time delay* of one sinusoid relative to its reference sinusoid.

The **instantaneous power** dissipated by any element is the product of its instantaneous voltage and current.

$$p(t) = v(t)i(t) = VI \cos(\omega t + \theta_V) \cos(\omega t + \theta_I) \quad (7.2)$$

This expression is further simplified with the aid of the trigonometric identity:

$$2 \cos(x) \cos(y) = \cos(x + y) + \cos(x - y) \quad (7.3)$$

Let $x = \omega t + \theta_V$ and $y = \omega t + \theta_I$ to yield:

$$\begin{aligned} p(t) &= \frac{VI}{2} [\cos(2\omega t + \theta_V + \theta_I) + \cos(\theta_V - \theta_I)] \\ &= \frac{VI}{2} [\cos(2\omega t + \theta_V + \theta_I) + \cos(\theta_Z)] \end{aligned} \quad (7.4)$$

Equation 7.4 illustrates that the total instantaneous power dissipated by an element is equal to the sum of a constant $\frac{1}{2} VI \cos(\theta_Z)$ and a sinusoidal $\frac{1}{2} VI \cos(2\omega t + \theta_V + \theta_I)$,

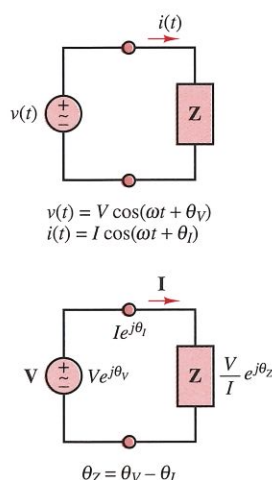


Figure 7.1 Time and frequency domain representations of an AC circuit. The phase angle of the load is $\theta_Z = \theta_V - \theta_I$.

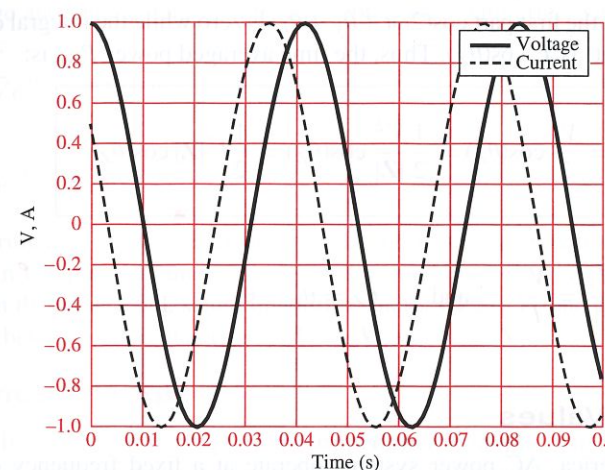


Figure 7.2 Current and voltage waveforms with unit amplitude and a phase shift of 60° .

which oscillates at twice the frequency of the source. Since the time average of a sinusoid is zero over one period or over a sufficiently long interval, the constant $\frac{1}{2} VI \cos(\theta_Z)$ is the time averaged power dissipated by a complex load Z , where θ_Z is the phase angle of that load.

Figure 7.3 shows the instantaneous and average power corresponding to the voltage and current signals of Figure 7.2. These observations can be confirmed mathematically by noting that the time average of the instantaneous power is defined by:

$$P_{\text{avg}} \equiv \frac{1}{T} \int_{t_0}^{t_0+T} P(t) dt \quad (7.5)$$

where T is one period of $p(t)$. Use equation 7.4 to substitute for $p(t)$ and yield:

$$\begin{aligned} P_{\text{avg}} &= \frac{1}{T} \int_{t_0}^{t_0+T} \frac{VI}{2} [\cos(2\omega t + \theta_V + \theta_I) + \cos(\theta_Z)] dt \\ &= \frac{VI}{2T} \int_{t_0}^{t_0+T} [\cos(2\omega t + \theta_V + \theta_I) + \cos(\theta_Z)] dt \end{aligned} \quad (7.6)$$

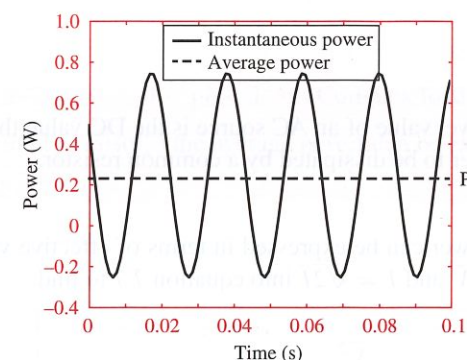


Figure 7.3 Instantaneous and average power corresponding to the signals in Figure 7.2

The integral of the first part $\cos(2\omega t + \theta_V + \theta_I)$ is zero while the integral of the second part (a constant) is $T \cos(\theta_Z)$. Thus, the time averaged power P_{avg} is:



$$P_{\text{avg}} = \frac{VI}{2} \cos(\theta_Z) = \frac{1}{2} \frac{V^2}{|Z|} \cos(\theta_Z) = \frac{1}{2} I^2 |Z| \cos(\theta_Z) \quad (7.7)$$

where

$$|Z| = \frac{|V|}{|I|} = \frac{V}{I} \quad \text{and} \quad \theta_Z = \theta_V - \theta_I \quad (7.8)$$

Effective Values

In North America, AC power systems operate at a fixed frequency of 60 cycles per second, or hertz (Hz), which corresponds to an angular (radian) frequency ω given by:

$$\omega = 2\pi \cdot 60 = 377 \text{ rad/s} \quad \text{AC power frequency} \quad (7.9)$$

In Europe and most other parts of the world, the AC power frequency is 50 Hz.



Unless indicated otherwise, the angular (radian) frequency ω is assumed to be 377 rad/s throughout this chapter.

It is customary in AC power analysis to employ the *effective* or *root-mean-square* (rms) amplitude (see section 4.2) rather than the peak amplitude for AC voltages and currents. In the case of a sinusoidal waveform, the effective voltage $\tilde{V} \equiv V_{\text{rms}}$ is related to the peak voltage V by:

$$\tilde{V} = V_{\text{rms}} = \frac{V}{\sqrt{2}} \quad (7.10)$$

Likewise, the effective current $\tilde{I} \equiv I_{\text{rms}}$ is related to the peak current I by:

$$\tilde{I} = I_{\text{rms}} = \frac{I}{\sqrt{2}} \quad (7.11)$$



The rms, or effective, value of an AC source is the DC value that produces the same average power to be dissipated by a common resistor.

The average power can be expressed in terms of effective voltage and current by plugging $V = \sqrt{2}\tilde{V}$ and $I = \sqrt{2}\tilde{I}$ into equation 7.7 to find:



$$P_{\text{avg}} = \tilde{V}\tilde{I} \cos(\theta_Z) = \frac{\tilde{V}^2}{|Z|} \cos(\theta_Z) = \tilde{I}^2 |Z| \cos(\theta_Z) \quad (7.12)$$

Voltage and current phasors are also represented with effective amplitudes by the notation:

$$\tilde{\mathbf{V}} = \tilde{V} e^{j\theta_V} = \tilde{V} \angle \theta_V \quad (7.13)$$

and

$$\tilde{\mathbf{I}} = \tilde{I} e^{j\theta_I} = \tilde{I} \angle \theta_I \quad (7.14)$$

It is critical to pay close attention to the *mathematical notation* that was first introduced in Chapter 4, namely that complex quantities, such as $\mathbf{V}$, $\mathbf{I}$, and $\mathbf{Z}$ are boldface. On the other hand, scalar quantities, such as V , I , $\tilde{V}$, and $\tilde{I}$ are italic. The relationship between these quantities is $V = |\mathbf{V}|$ and $\tilde{V} = |\tilde{\mathbf{V}}|$.

Impedance Triangle

Figure 7.4 illustrates the concept of the **impedance triangle**, which is an important graphical representation of impedance as a vector in the complex plane. Basic trigonometry yields:

$$R = |Z| \cos \theta \quad (7.15)$$

$$X = |Z| \sin \theta \quad (7.16)$$

where R is the *resistance* and X is the *reactance*. Notice that both R and P_{avg} are proportional to $\cos(\theta_Z)$, which suggests that a triangle similar to (i.e., the same shape as) the impedance triangle could be constructed with P_{avg} as one leg of a right triangle. In fact, such a triangle is known as a *power triangle*. The similarity of these two types of triangles is a powerful concept for problem solving, as is shown in Section 7.2.

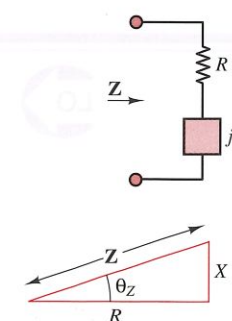


Figure 7.4 Impedance triangle

Power Factor

The phase angle θ_Z of the load impedance plays a very important role in AC power circuits. From equation 7.12, the average power dissipated by an AC load is proportional to $\cos(\theta_Z)$. For this reason, $\cos(\theta_Z)$ is known as the **power factor (pf)**. For purely resistive loads:

$$\theta_Z = 0 \rightarrow \text{pf} = 1 \quad \text{Resistive load} \quad (7.17)$$

For purely inductive or capacitive loads:

$$\theta_Z = +\pi/2 \rightarrow \text{pf} = 0 \quad \text{Inductive load} \quad (7.18)$$

$$\theta_Z = -\pi/2 \rightarrow \text{pf} = 0 \quad \text{Capacitive load} \quad (7.19)$$

For loads with non-zero resistive (real) and reactive (imaginary) parts:

$$0 < |\theta_Z| < \pi/2 \rightarrow 0 < \text{pf} < 1 \quad \text{Complex load} \quad (7.20)$$

Using the definition $\text{pf} = \cos(\theta_Z)$ the average power can be expressed as:

$$P_{\text{avg}} = \tilde{V}\tilde{I} \text{pf} \quad (7.21)$$

Thus, average power dissipated by a resistor is:

$$(P_{\text{avg}})_R = \tilde{V}_R \tilde{I}_R \text{pf}_R = \tilde{V}_R \tilde{I}_R \quad (7.22)$$

because $\text{pf}_R = 1$. By contrast, the average power dissipated by a capacitor or inductor is:

$$(P_{\text{avg}})_X = \tilde{V}_X \tilde{I}_X \text{pf}_X = 0 \quad (7.23)$$

because $\text{pf}_X = 0$, where the subscript X indicates a reactive element (i.e., either a capacitor or inductor). It is important to note that although capacitors and inductors are *lossless* (i.e., they store and release energy but do not dissipate energy), they do influence power dissipation in a circuit by affecting the voltage across and the current through resistors in the circuit.

When θ_Z is positive, the load is *inductive* and the power factor is said to be *lagging*; when θ_Z is negative, the load is *capacitive* and the power factor is said to be *leading*. It is important to keep in mind that $\text{pf} = \cos(\theta_Z) = \cos(-\theta_Z)$ because the cosine is an even function. Thus, while it may be important to know whether a load is inductive or capacitive, the value of the power factor only indicates the extent to which a load is inductive or capacitive. To know whether a load is inductive or capacitive, one must know whether the power factor is leading or lagging.



EXAMPLE 7.1 Computing Average and Instantaneous AC Power

Problem

Compute the average and instantaneous power dissipated by the load of Figure 7.5.

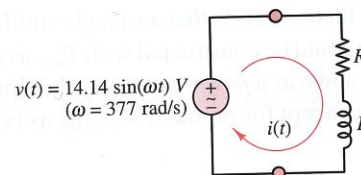


Figure 7.5

Solution

Known Quantities: Source voltage and frequency, load resistance and inductance values.

Find: P_{avg} and $p(t)$ for the RL load.

Schematics, Diagrams, Circuits, and Given Data: $v(t) = 14.14 \sin(377t) \text{ V}$; $R = 4 \Omega$; $L = 8 \text{ mH}$.

Assumptions: None.

Analysis: The source voltage is expressed in terms of $\sin(377t)$. By convention, all time-domain sinusoids should be expressed as cosines. To convert $\sin(377t)$ to $\cos(377t + \theta_V)$ recall that a sine equals a cosine shifted forward in time (to the right) by $\pi/2$ rad; that is, $\sin(377t) = \cos(377t - \pi/2)$. Thus, at the angular frequency $\omega = 377 \text{ rad/s}$ the source voltage is:

$$\tilde{V} = 10 \angle \left(-\frac{\pi}{2}\right) \text{ V rms}$$

where $14.14 \text{ V} = 10 \text{ V rms}$.

The equivalent impedance of the load is:

$$Z = R + j\omega L \approx 4 + j3 = 5 \angle (36.9^\circ) = 5 \angle (0.644 \text{ rad}) \Omega$$

The current in the loop is:

$$\tilde{I} = \frac{\tilde{V}}{Z} \approx \frac{10 \angle (-\pi/2)}{5 \angle (0.644)} \approx 2 \angle (-2.215) \text{ A rms}$$

It is instructive to compute the average power dissipated in the circuit in two ways:

1. The most straightforward and brute force approach is to compute:

$$P_{\text{avg}} = \tilde{V} \tilde{I} \cos(\theta_Z) = 10 \times 2 \times \cos(0.644) \approx 16 \text{ W}$$

2. Another approach is to realize that the average power dissipated by the inductor is zero. Thus, the total average power dissipated equals the average power dissipated by the resistor. Thus:

$$(P_{\text{avg}})_R = \tilde{I}^2 R \text{pf}_R = \tilde{I}^2 R \approx (2)^2 \times 4 = 16 \text{ W}$$

The instantaneous power is given by:

$$p(t) = v(t) \times i(t) = \sqrt{2} \times 10 \sin(377t) \times \sqrt{2} \times 2 \cos(377t - 2.215) \text{ W}$$

The instantaneous voltage and current waveforms and the instantaneous and average power are plotted in Figure 7.6.

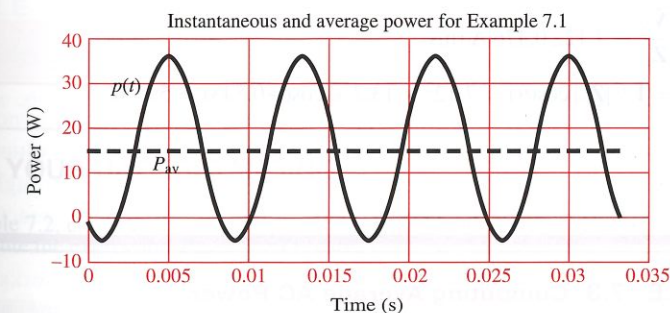
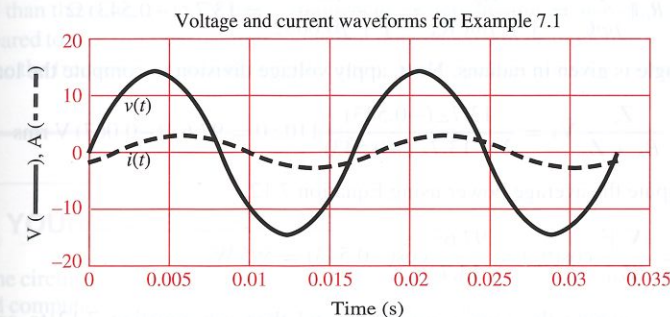


Figure 7.6

Comment: It is standard procedure in electrical engineering practice to use rms values in power calculations. Also, note that the instantaneous power can be negative at times even though the average power is positive. This result reflects the fact that although the average power of an inductor is identically zero, the instantaneous power of an inductor can be positive or negative as the inductor charges or discharges with the sinusoidal source.



EXAMPLE 7.2 Computing Average AC Power

Problem

Compute the average power dissipated by the load of Figure 7.7.

Solution

Known Quantities: Source voltage, internal resistance, load resistance, capacitance, and frequency.

Find: P_{avg} for the $R_o \parallel C_o$ load.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_s = 110\angle 0^\circ$ V rms; $R_s = 2\ \Omega$; $R_o = 16\ \Omega$; $C = 100\ \mu\text{F}$; $\omega = 377$ rad/s.

Assumptions: None.

Analysis: First, compute the impedance of the load at the angular frequency $\omega = 377$ rad/s:

$$\mathbf{Z}_o = R_o \parallel \frac{1}{j\omega C} = \frac{R_o}{1 + j\omega C R_o} = \frac{16}{1 + j0.6032} = 13.7\angle(-0.543)\ \Omega$$

where the angle is given in radians. Next, apply voltage division to compute the load voltage:

$$\tilde{V}_o = \frac{\mathbf{Z}_o}{R_s + \mathbf{Z}_o} \tilde{V}_s = \frac{13.7\angle(-0.543)}{2 + 13.7\angle(-0.543)} 110\angle 0 = 97.6\angle(-0.067)\ \text{V rms}$$

Finally, compute the average power using Equation 7.12:

$$P_{\text{avg}} = \frac{|\tilde{V}_o|^2}{|\mathbf{Z}_o|} \cos(\theta_Z) = \frac{97.6^2}{13.7} \cos(-0.543) = 595\ \text{W}$$

Alternatively, compute the source current $\tilde{\mathbf{I}}_s$ and then use equation 7.12 to compute the average power:

$$\tilde{\mathbf{I}}_s = \frac{\tilde{V}_o}{\mathbf{Z}_o} = 7.12\angle 0.476\ \text{A rms}$$

$$P_{\text{avg}} = |\tilde{\mathbf{I}}_s|^2 |\mathbf{Z}_o| \cos(\theta) = 7.12^2 \times 13.7 \times \cos(-0.543) = 595\ \text{W}$$



EXAMPLE 7.3 Computing Average AC Power

Problem

Compute the average power dissipated by the load of Figure 7.8.

Solution

Known Quantities: Source voltage, internal resistance, load resistance, capacitance and inductance values, and frequency.

Find: P_{avg} for the complex load.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_s = 110\angle 0^\circ$ V rms; $R = 10\ \Omega$; $L = 0.05\text{H}$; $C = 470\ \mu\text{F}$; $\omega = 377$ rad/s. Figure 7.8.

Assumptions: None.

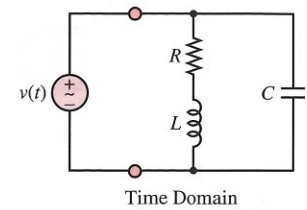
Analysis: First, compute the impedance of the load $\mathbf{Z}_o$ at the angular frequency $\omega = 377$ rad/s:

$$\begin{aligned} \mathbf{Z}_o &= (R + j\omega L) \parallel \frac{1}{j\omega C} = \frac{(R + j\omega L)/j\omega C}{R + j\omega L + 1/j\omega C} \\ &= \frac{R + j\omega L}{1 - \omega^2 LC + j\omega CR} = 1.16 - j7.18 \\ &= 7.27\angle(-1.41)\ \Omega \end{aligned}$$

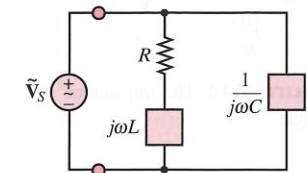
Note that the equivalent load impedance at $\omega = 377$ rad/s has a negative imaginary part, which is a feature of a *capacitive load*, as shown in Figure 7.9. The average power is:

$$P_{\text{avg}} = \frac{|\tilde{V}_s|^2}{|\mathbf{Z}_o|} \cos(\theta) = \frac{110^2}{7.27} \cos(-1.41) = 266\ \text{W}$$

Comment: At $\omega = 377$ rad/s, the capacitance has a larger impact on the total equivalent impedance than the inductance. At lower frequencies, where the impedance of the capacitor is large compared to $R + j\omega L$, the parallel equivalent impedance will be inductive. It is instructive to determine the frequencies when the parallel equivalent impedance has a zero imaginary part.



Time Domain



Frequency Domain

Figure 7.8

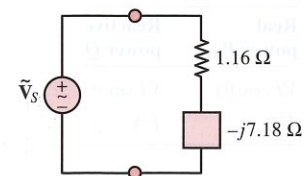


Figure 7.9

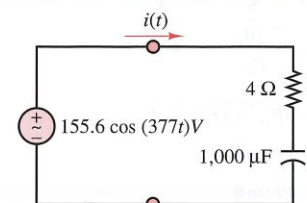


Figure 7.10

CHECK YOUR UNDERSTANDING

Consider the circuit shown in Figure 7.10. Find the impedance of the load “seen” by the voltage source, and compute the average power dissipated by the load. The constant 155.6 multiplying the cosine function is always the peak amplitude, not the rms amplitude.

Answer: $\mathbf{Z} = 4.8e^{-j33.5^\circ}\ \Omega$; $P_{\text{avg}} = 2,103.4\ \text{W}$

CHECK YOUR UNDERSTANDING

For Example 7.2, compute the average power dissipated by the internal source resistance R_s . Also compute the expression $P_{\text{avg}} = I_s^2 |\mathbf{Z}| \cos(\theta_Z)$.

Answers: 101.46 W; 595 W

7.2 COMPLEX POWER

The computation of AC power is simplified by defining a **complex power S**, where:

$$\mathbf{S} = \tilde{\mathbf{V}} \tilde{\mathbf{I}}^* \quad \text{Complex power} \quad (7.24)$$



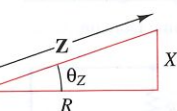
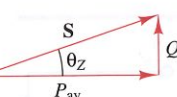


Figure 7.11 The impedance triangle

Table 7.1 Real and reactive power

Real power P_{avg}	Reactive power Q
$\tilde{V}\tilde{I} \cos(\theta)$	$\tilde{V}\tilde{I} \sin(\theta)$
$\tilde{I}^2 R$	$\tilde{I}^2 X$



$$S = \sqrt{P_{\text{av}}^2 + Q^2} = \tilde{V} \cdot \tilde{I}$$

$$P_{\text{av}} = \tilde{V}\tilde{I} \cos \theta$$

$$Q = \tilde{V}\tilde{I} \sin \theta$$

Figure 7.12 The complex power triangle

where the asterisk denotes the complex conjugate (see Appendix A). Note that the effect of taking the complex conjugate of a phasor is to multiply its phase angle by -1 . In other words, $\angle \mathbf{S} = \angle \tilde{\mathbf{V}} + \angle \tilde{\mathbf{I}}^* = \angle \tilde{\mathbf{V}} - \angle \tilde{\mathbf{I}} = \theta_Z$. The definition of complex power leads to:

$$\begin{aligned} \mathbf{S} &= \tilde{V}\tilde{I} \cos(\theta_Z) + j\tilde{V}\tilde{I} \sin(\theta_Z) \\ &= \tilde{I}^2 |\mathbf{Z}| \cos(\theta_Z) + j\tilde{I}^2 |\mathbf{Z}| \sin(\theta_Z) \\ &= \tilde{I}^2 R + j\tilde{I}^2 X = \tilde{I}^2 \mathbf{Z} \end{aligned} \quad (7.25)$$

where $R = |\mathbf{Z}| \cos(\theta_Z)$ and $X = |\mathbf{Z}| \sin(\theta_Z)$ are the resistance and reactance of the impedance triangle shown in Figure 7.11. The real and imaginary parts of $\mathbf{S}$ are the **real power** $P_{\text{avg}} = \tilde{V}\tilde{I} \cos(\theta_Z)$ and the **reactive power** $Q = \tilde{V}\tilde{I} \sin(\theta_Z)$, respectively, such that:

$$\mathbf{S} = P_{\text{avg}} + jQ \quad (7.26)$$

The magnitude $|\mathbf{S}|$ of the complex power is called the **apparent power** S and is measured in units of **volt-amperes (VA)**. The units of Q are **volt-amperes reactive**, or **VAR**.

The relationship between S , P , and Q is summarized by a **power triangle** as shown in Figure 7.12. It is important to note that the impedance and power triangles are similar; that is, they have the same shape. This result is helpful in problem solving. Table 7.1 shows the general expressions for calculating P and Q .

The complex power can also be expressed as:

$$\mathbf{S} = \tilde{I}^2 \mathbf{Z} = \tilde{I}^2 R + j\tilde{I}^2 X \quad (7.27)$$

Furthermore, since $\tilde{V} = \tilde{I}|\mathbf{Z}|$ and $|\mathbf{Z}|^2 = \mathbf{Z}\mathbf{Z}^*$, the complex power can be re-expressed as:

$$\begin{aligned} \mathbf{S} &= \tilde{I}^2 \mathbf{Z} = \frac{\tilde{I}^2 \mathbf{Z} \mathbf{Z}^*}{\mathbf{Z}^*} \\ &= \frac{\tilde{V}^2}{\mathbf{Z}^*} \end{aligned} \quad (7.28)$$

As previously stated, capacitors and inductors (reactive loads) do not dissipate energy themselves; they are lossless elements. However, they do influence power dissipation in a circuit by affecting the voltage across and current through resistors, which do dissipate energy. This influence is now quantified by the reactive power, Q , which is due entirely to capacitance and inductance in a circuit. It is worth noting that $Q = 0$, $\text{pf} = 1$, and therefore $P = S$ in purely resistive networks. It is also important to realize that P represents the real work done (per unit time) by a circuit. For example, the real power P of an electric motor represents the work done (per unit time) by the motor to perform some useful task. From the perspective of the utility company that provides the electric power for the motor and of the owner of the motor who has to pay the utility bill, it would be best if all the apparent power S provided by the utility company was converted to useful power P . (Why?) However, all electric motors have some inductance (e.g., coils of wire) such that $Q \neq 0$, $\text{pf} < 1$, and $P < S$. It is possible to *correct* the effect of a motor's inductance by adding capacitance in parallel with the motor so as to decrease Q and thereby decrease the apparent power S that must be provided for a given P required by the task.

FOCUS ON PROBLEM SOLVING

COMPLEX POWER COMPUTATION

1. Use AC circuit analysis methods to compute (as phasors) the voltage across and current through the load. Convert peak amplitudes to effective (rms) values.

$$\tilde{\mathbf{V}} = \tilde{V} \angle \theta_V \quad \text{and} \quad \tilde{\mathbf{I}} = \tilde{I} \angle \theta_I$$

2. Compute $\theta_Z = \theta_V - \theta_I$ and the power factor $\text{pf} = \cos(\theta_Z)$. Draw the impedance triangle, as shown in Figure 7.11.
3. Use one of the two following methods to compute P_{avg} and Q .
 - Compute the complex power $\mathbf{S} = \tilde{\mathbf{V}}\tilde{\mathbf{I}}^*$ such that $P = P_{\text{avg}} = \text{Re}(\mathbf{S})$, $Q = \text{Im}(\mathbf{S})$, and $S = |\mathbf{S}|$. The effect of taking the complex conjugate of a phasor is to multiply its phase angle by -1 , such that $\angle \mathbf{S} = \angle \tilde{\mathbf{V}} - \angle \tilde{\mathbf{I}} = \theta_Z$.
 - Compute the apparent power $S = |\mathbf{S}| = \tilde{V}\tilde{I}$ such that $P = P_{\text{avg}} = S \text{pf}$ and $Q = S \sin(\theta_Z)$.
4. Draw the power triangle, as shown in Figure 7.12, and confirm that $S^2 = P^2 + Q^2$ and that $\tan(\theta_Z) = Q/P$.
5. If Q is negative, the load is capacitive and the power factor is *leading*; if Q is positive, the load is inductive and the power factor is *lagging*.

EXAMPLE 7.4 Complex Power Calculations

Problem

Compute the complex power for the load $\mathbf{Z}_o$ of Figure 7.13.

Solution

Known Quantities: Source, load voltage, and current.

Find: $\mathbf{S} = P_{\text{avg}} + jQ$ for the complex load.

Schematics, Diagrams, Circuits, and Given Data: $v(t) = 100 \cos(\omega t + 0.262) \text{ V}$; $i(t) = 2 \cos(\omega t - 0.262) \text{ A}$; $\omega = 377 \text{ rad/s}$.

Assumptions: All angles are given in units of radians unless indicated otherwise.

Analysis: First, realize that the constants multiplying the cosine functions are always peak, not rms, values. These functions can be converted to phasor quantities with rms amplitudes as follows:

$$\tilde{\mathbf{V}} = \frac{100}{\sqrt{2}} \angle 0.262 \text{ V} \quad \tilde{\mathbf{I}} = \frac{2}{\sqrt{2}} \angle (-0.262) \text{ A}$$

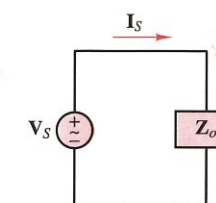


Figure 7.13

Compute the phase angle of the load, and the real and reactive power, using the definitions of equation 7.12:

$$\theta_Z = \angle(\tilde{V}) - \angle(\tilde{I}) = 0.524 \text{ rad}$$

$$P_{\text{avg}} = |\tilde{V}||\tilde{I}| \cos(\theta_Z) = \frac{200}{2} \cos(0.524) = 86.6 \text{ W}$$

$$Q = |\tilde{V}||\tilde{I}| \sin(\theta_Z) = \frac{200}{2} \sin(0.524) = 50 \text{ VAR}$$

Apply the definition of complex power (equation 7.24) to repeat the same calculation:

$$S = \tilde{V}\tilde{I}^* = \frac{100}{\sqrt{2}} \angle 0.262 \times \frac{2}{\sqrt{2}} \angle -(-0.262) = 100 \angle 0.524 \\ = (86.6 + j50) \text{ VA}$$

Therefore

$$P_{\text{avg}} = 86.6 \text{ W} \quad Q = 50 \text{ VAR}$$

Comments: Note how the definition of complex power yields both quantities at one time.



EXAMPLE 7.5 Real and Reactive Power Calculations

Problem

Compute the complex power for the load of Figure 7.14.

Solution

Known Quantities: Source voltage and resistance; load impedance.

Find: $S = P + jQ$ for the complex load.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_S = 110 \angle 0^\circ \text{ V}$; $R_S = 2 \Omega$; $R = 5 \Omega$; $C = 2,000 \mu\text{F}$; $\omega = 377 \text{ rad/s}$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise.

Analysis: The load impedance is:

$$Z_o = R + \frac{1}{j\omega C} = (5 - j1.326) \Omega = 5.173 \angle (-0.259) \Omega$$

Next, apply voltage division and the generalized Ohm's law to compute the load voltage and current:

$$\tilde{V}_o = \frac{Z_o}{R_S + Z_o} \tilde{V}_S = \frac{5 - j1.326}{7 - j1.326} \times 110 = 79.86 \angle (-0.072) \text{ V}$$

$$\tilde{I}_o = \frac{\tilde{V}_o}{Z_o} = \frac{79.86 \angle (-0.072)}{5.173 \angle (-0.259)} = 15.44 \angle 0.187 \text{ A}$$

Finally, compute the complex power, as defined in equation 7.24:

$$S = \tilde{V}_o \tilde{I}_o^* = 79.9 \angle (-0.072) \times 15.44 \angle (-0.187) = 1,233 \angle (-0.259) \\ = (1,192 - j316) \text{ VA}$$

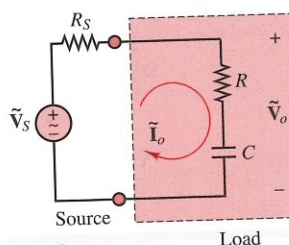


Figure 7.14

Therefore:

$$P = 1,192 \text{ W} \quad Q = -316 \text{ VAR}$$

Comment: Is the reactive power capacitive or inductive? Since $Q < 0$, the reactive power is capacitive!

EXAMPLE 7.6 Real Power Transfer for Complex Loads

Problem

Compute the complex power for the load between terminals a and b of Figure 7.15. Repeat the computation with the inductor removed from the load, and compare the real power for the two cases.

Solution

Known Quantities: Source voltage and resistance; load impedance.

Find:

1. $S_1 = P_1 + jQ_1$ for the complex load.
2. $S_2 = P_2 + jQ_2$ for the real load.
3. For each case, compute the ratio of the real power dissipated by the load to the overall real power dissipated by the circuit.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_S = 110 \angle 0^\circ \text{ V}$; $R_S = 4 \Omega$; $R = 10 \Omega$; $jX_L = j6 \Omega$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise.

Analysis:

1. With the inductor included in the load, its impedance Z_o is:

$$Z_o = R \parallel j\omega L = \frac{10 \times j6}{10 + j6} = 5.145 \angle 1.03 \Omega$$

Apply voltage division to compute the load voltage $\tilde{V}_o$ and the generalized Ohm's law to compute the current $\tilde{I}_o = \tilde{I}_S$.

$$\tilde{V}_o = \frac{Z_o}{R_S + Z_o} \tilde{V}_S = \frac{5.145 \angle 1.03}{4 + 5.145 \angle 1.03} \times 110 = 70.9 \angle 0.444 \text{ V}$$

$$\tilde{I}_o = \frac{\tilde{V}_o}{Z_o} = \frac{70.9 \angle 0.444}{5.145 \angle 1.03} = 13.8 \angle (-0.586) \text{ A}$$

Finally, compute the complex power, as defined in equation 7.24:

$$S_1 = \tilde{V}_o \tilde{I}_o^* = 70.9 \angle 0.444 \times 13.8 \angle 0.586 = 978 \angle 1.03 \text{ VA} \\ = (503 + j839) \text{ VA}$$

Therefore:

$$P_1 = 503 \text{ W} \quad Q_1 = 839 \text{ VAR}$$

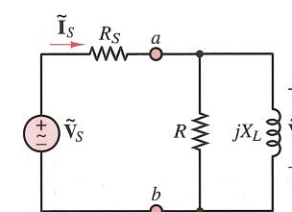


Figure 7.15

2. With the inductor excluded from the load (Figure 7.16), its impedance is:

$$Z_o = R = 10 \Omega$$

Compute the load voltage and current:

$$\tilde{V}_o = \frac{Z_o}{R_S + Z_o} \tilde{V}_S = \frac{10}{4 + 10} \times 110 = 78.57 \angle 0^\circ \text{ V}$$

$$\tilde{I}_o = \frac{\tilde{V}_o}{Z_o} = \frac{78.57 \angle 0^\circ}{10} = 7.857 \angle 0^\circ \text{ A}$$

Finally, compute the complex power, as defined in equation 7.24:

$$S_2 = \tilde{V}_o \tilde{I}_o^* = 78.57 \angle 0^\circ \times 7.857 \angle 0^\circ = 617 \angle 0^\circ = (617 + j0) \text{ VA}$$

Therefore:

$$P_2 = 617 \text{ W} \quad Q_2 = 0 \text{ VAR}$$

3. To compute the overall real power P_{total} dissipated by the circuit, it is necessary to include the impact of the line resistance R_S and compute for each case:

$$S_{\text{total}} = \tilde{V}_S \tilde{I}_S^* = P_{\text{total}} + jQ_{\text{total}}$$

For case 1:

$$\tilde{I}_S = \frac{\tilde{V}_S}{Z_{\text{total}}} = \frac{\tilde{V}_S}{R_S + Z_o} = \frac{110}{4 + 5.145 \angle 1.03} = 13.8 \angle (-0.586) \text{ A}$$

$$S_{1\text{total}} = \tilde{V}_S \tilde{I}_S^* = 110 \times 13.8 \angle (+0.586) = (1,264 + j838) \text{ VA} = P_{1\text{total}} + jQ_{1\text{total}}$$

The percent real power transfer is:

$$100 \times \frac{P_1}{P_{1\text{total}}} = \frac{503}{1,264} = 39.8\%$$

For case 2:

$$\tilde{I}_S = \frac{\tilde{V}_S}{Z_{\text{total}}} = \frac{\tilde{V}_S}{R_S + R} = \frac{110}{4 + 10} = 7.857 \angle 0^\circ \text{ A}$$

$$S_{2\text{total}} = \tilde{V}_S \tilde{I}_S^* = 110 \times 7.857 = (864 + j0) \text{ VA} = P_{2\text{total}} + jQ_{2\text{total}}$$

The percent real power transfer is:

$$100 \times \frac{P_2}{P_{2\text{total}}} = \frac{617}{864} = 71.4\%$$

Comments: If it were possible to eliminate the reactive part of the impedance, the percentage of real power transferred from the source to the load would be increased significantly. The procedure to accomplish this goal is called *power factor correction*.



EXAMPLE 7.7 Complex Power and Power Triangle

Problem

Find the reactive and real power for the load of Figure 7.17. Draw the associated power triangle.

Solution

Known Quantities: Source voltage; load impedance.

Find: $S = P_{\text{avg}} + jQ$ for the complex load.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_S = 60 \angle 0^\circ \text{ V}$; $R = 3 \Omega$; $jX_L = j9 \Omega$; $jX_C = -j5 \Omega$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians.

Analysis: First, compute the load current:

$$\tilde{I}_o = \frac{\tilde{V}_o}{Z_o} = \frac{60 \angle 0^\circ}{3 + j9 - j5} = \frac{60 \angle 0^\circ}{5 \angle 0.9273} = 12 \angle (-0.9273) \text{ A}$$

Next, compute the complex power, as defined in equation 7.24:

$$S = \tilde{V}_o \tilde{I}_o^* = 60 \angle 0^\circ \times 12 \angle 0.9273 = 720 \angle 0.9273 = (432 + j576) \text{ VA}$$

Therefore:

$$P = 432 \text{ W} \quad Q = 576 \text{ VAR}$$

The total reactive power must be the sum of the reactive powers in each of the elements, such that $Q = Q_C + Q_L$. Compute these two quantities as follows:

$$Q_C = |\tilde{I}_o|^2 \times X_C = (144)(-5) = -720 \text{ VAR}$$

$$Q_L = |\tilde{I}_o|^2 \times X_L = (144)(9) = 1,296 \text{ VAR}$$

and

$$Q = Q_L + Q_C = 576 \text{ VAR}$$

Comments: The power triangle corresponding to this circuit is drawn in Figure 7.18. The vector diagram shows how the complex power S results from the vector addition of the three components P , Q_C , and Q_L .

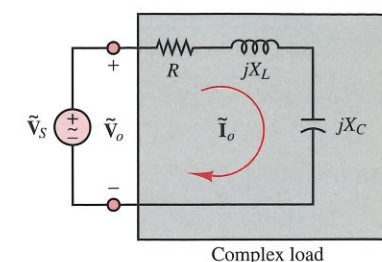
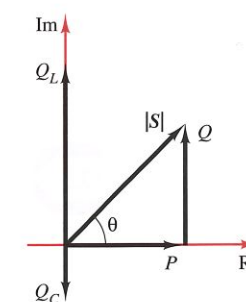


Figure 7.17



Note: $S = P + jQ_C + jQ_L$

Figure 7.18

CHECK YOUR UNDERSTANDING

Compute the real and reactive power for the load of Example 7.2.

$$\text{Answer: } P_{\text{avg}} = 595 \text{ W}; Q = -359 \text{ VAR}$$

CHECK YOUR UNDERSTANDING

Compute the real and reactive power for the load of Figure 7.10.

$$\text{Answer: } P_{\text{avg}} = 2.1 \text{ kW}; Q = 1.39 \text{ kVAR}$$

CHECK YOUR UNDERSTANDING

Refer to Example 7.6, and compute the percent of real power transfer for the case where the inductance of the load is one-half of the original value.

Answer: 29.3 %

CHECK YOUR UNDERSTANDING

Compute the power factor for the load of Example 7.7 with and without the inductor in the circuit.

Answer: pf = 0.6, lagging (with L in circuit); pf = 0.5145, leading (without L)

7.3 POWER FACTOR CORRECTION

A power factor close to unity signifies an efficient transfer of energy from the AC source to the load while a small power factor corresponds to inefficient use of energy, as illustrated in Example 7.6. If a load requires a given real power P , the current required by the load will be minimized when the power factor is maximized, that is, when $\text{pf} = \cos(\theta_Z) \rightarrow 1$. When $\text{pf} < 1$, it is possible to increase it (i.e., *correct* it) by adding, as appropriate, reactance (e.g., capacitance) to the load. When pf is leading, inductance must be added; when pf is lagging, capacitance must be added.

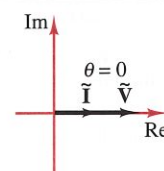
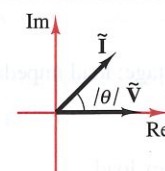
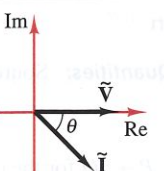
If $\theta_Z > 0$, then $Q > 0$, the load is inductive, the load current *lags* the load voltage, and the power factor pf is lagging. Alternatively, if $\theta_Z < 0$, then $Q < 0$, the load is capacitive, the load current *leads* the load voltage, and the power factor pf is leading.

Table 7.2 illustrates and summarizes these concepts. For simplicity, the phase angle of the voltage phasor $\tilde{V}$ shown in the table is zero and acts as a reference angle for the current phasor.

In practice, the load designed for a useful industrial task is often inductive because of the presence of electric motors. The power factor of an inductive load can be *corrected* by adding capacitance in parallel with the load. This procedure is called *power factor correction*.

The measurement and correction of the power factor for the load are an extremely important aspect of any industrial engineering application that requires the use of substantial quantities of electric power. In particular, industrial plants, construction sites, heavy machinery, and other heavy users of electric power must be aware of the power factor that their loads present to the electric utility company. As was already observed, a low power factor results in greater current draw from the electric utility and greater line losses. Thus, computations related to the power factor of complex loads are of great utility to any practicing engineer.

Table 7.2 Important facts related to complex power

	Resistive load	Capacitive load	Inductive load
Ohm's law	$\tilde{V} = \tilde{I}Z$	$\tilde{V} = \tilde{I}Z$	$\tilde{V} = \tilde{I}Z$
Complex impedance	$Z = R$	$Z = R + jX$ $X < 0$	$Z = R + jX$ $X > 0$
Phase angle	$\theta = 0$	$\theta < 0$	$\theta > 0$
Complex plane sketch			
Explanation	The current is in phase with the voltage.	The current "leads" the voltage.	The current "lags" the voltage.
Power factor	Unity	Leading, < 1	Lagging, < 1
Reactive power	0	Negative	Positive

FOCUS ON PROBLEM SOLVING

POWER FACTOR CORRECTION

- Follow the steps outlined in the Focus on Problem Solving box "Complex Power Computation" to find the initial phase angle of the load θ_{Z_i} , power factor pf_i , real power P_i , and reactive power Q_i . If both P_i and either pf or θ_Z are given, compute Q directly using $Q = P \tan(\theta_Z)$. An initial power triangle is helpful for visualizing this information.
- For a lagging power factor, augment the load with a parallel capacitor such that:

$$\Delta Q = Q_c = \frac{\tilde{V}^2}{|Z_c|} \sin(\theta_Z) = -\omega C \tilde{V}^2$$

- Express the final reactive power Q_f as:

$$Q_f = Q_i + \Delta Q$$

- The real power is unchanged by the addition of the capacitor in parallel. Thus, $P_f = P_i$ and the final (corrected) phase angle of the augmented load is:

$$\theta_{Z_f} = \tan^{-1} \left(\frac{Q_f}{P_f} \right)$$

It is helpful to draw a final power triangle to visualize the effect of the parallel capacitor.

- The final corrected power factor is:

$$\text{pf}_f = \cos(\theta_{Z_f})$$



EXAMPLE 7.8 Power Factor Correction

Problem

Calculate the power factor for the circuit of Figure 7.19. Correct it to unity by adding a capacitor in parallel with the load.

Solution

Known Quantities: Source voltage; load impedance.

Find:

1. $S = P + jQ$ for the complex load.
2. Value of parallel capacitance that results in $\text{pf} = 1$.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_S = 117\angle 0^\circ \text{ V rms}$; $R = 50\ \Omega$; $jX_L = j86.7\ \Omega$; $\omega = 377 \text{ rad/s}$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise.

Analysis:

1. First, compute the load impedance:

$$Z_o = R + jX_L = 50 + j86.7 = 100\angle 1.047^\circ \Omega$$

Next, compute the load current $\tilde{I}_o = \tilde{I}_S$:

$$\tilde{I}_o = \frac{\tilde{V}_o}{Z_o} = \frac{117\angle 0^\circ}{50 + j86.7} = \frac{117\angle 0^\circ}{100\angle 1.047^\circ} = 1.17\angle (-1.047^\circ) \text{ A}$$

The complex power, as defined in equation 7.24, is:

$$S = \tilde{V}_o \tilde{I}_o^* = 117\angle 0^\circ \times 1.17\angle 1.047^\circ = 137\angle 1.047^\circ = (68.4 + j118.5) \text{ VA}$$

Therefore:

$$P = 68.4 \text{ W} \quad Q = 118.5 \text{ VAR}$$

The power triangle corresponding to this circuit is drawn in Figure 7.20. The vector diagram shows how the complex power S results from the vector addition of the two components P and Q .

2. To correct the power factor to unity it is necessary to subtract 118.5 VAR. This goal can be accomplished by adding in parallel a capacitor with $Q_C = -118.5 \text{ VAR}$. The required capacitance is found by:

$$X_C = \frac{|\tilde{V}_o|^2}{Q_C} = -\frac{(117)^2}{118.5} = -115 \Omega$$

The reactance X_C is related to the capacitance by:

$$jX_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

Thus, the result is:

$$C = -\frac{1}{\omega X_C} = -\frac{1}{377(-115)} = 23.1 \mu\text{F}$$

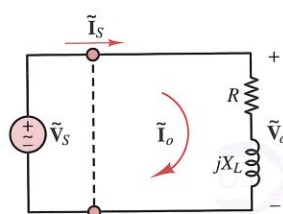


Figure 7.19

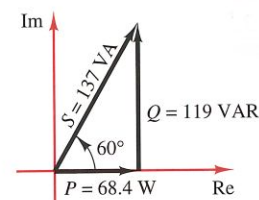


Figure 7.20

3. The total current required of the source is $\tilde{I}_S = \tilde{I}_o + \tilde{I}_C$, where:

$$\tilde{I}_C = \frac{\tilde{V}_S}{Z_C} = (j\omega C)(117\angle 0^\circ) = (377)(23.1 \mu\text{F})(117\angle (\pi/2)) \approx 1.02\angle 90^\circ \text{ A}$$

Notice that $|\tilde{I}_C| = |\tilde{V}_S|/|X_C| \approx 117/115 \approx 1.02 \text{ A}$. The total current is computed by phasor addition to be:

$$\tilde{I}_S \approx 1.17\angle (-1.047^\circ) + 1.02\angle (\pi/2) \approx 0.585\angle 0^\circ \text{ A}$$

The corrected power factor $\text{pf} = 1$ implies that the impedance of the load is now purely real; that is, $\theta_Z = 0$. Thus, the source current must now be *in phase* with the source voltage; and it is.

Comments: Notice that the magnitude of the source current is reduced by increasing the power factor. The power factor correction, which is a very common procedure in electric power systems, is illustrated in Figure 7.21.

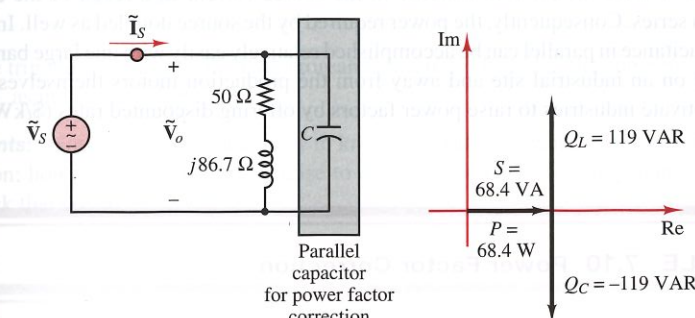


Figure 7.21 Power factor correction

EXAMPLE 7.9 Can a Series Capacitor Be Used for Power Factor Correction?

Problem

The circuit of Figure 7.22 suggests the use of a series capacitor for power factor correction. Why is this approach *not* a feasible alternative to the parallel capacitor approach demonstrated in Example 7.8?

Solution

Known Quantities: Source voltage; load impedance.

Find: Load (source) current.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_S = 117\angle 0^\circ \text{ V}$; $R = 50\ \Omega$; $jX_L = j86.7\ \Omega$; $jX_C = -j86.7\ \Omega$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise.

Analysis: First, compute the impedance of the load between terminals a and b :

$$Z_o = R + jX_L + jX_C = 50 + j86.7 - j86.7 = 50 \Omega$$

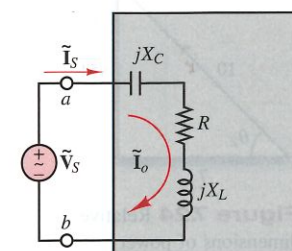


Figure 7.22

Notice that the reactance of the capacitor was chosen so as to make the total load purely resistive. Thus, $\theta_Z = 0$ and the corrected power factor is $\text{pf} = 1$. So far, so good.

Next, compute the current through the series load:

$$\tilde{I}_o = \tilde{I}_s = \frac{\tilde{V}_s}{Z_o} = \frac{117\angle 0}{50} = 2.34 \text{ A}$$

The corrected power factor $\text{pf} = 1$ implies that the impedance of the load is now purely real; that is, $\theta_Z = 0$. Thus, the source current must now be *in phase* with the source voltage; and it is.

The problem with this approach to power factor correction is revealed by computing the initial current through the load, prior to the addition of the capacitor.

$$(\tilde{I}_o)_{\text{initial}} = \frac{\tilde{V}_s}{R + jX_L} = \frac{117\angle 0}{50 + j86.7} \approx 1.17\angle(-\pi/3) \text{ A}$$

Comments: Notice the twofold increase in the source current as a result of the additional capacitor in series. Consequently, the power required by the source doubled as well. In practice, adding capacitance in parallel can be accomplished relatively easily with one large bank located somewhere on an industrial site and away from the production motors themselves. Electric utilities motivate industries to raise power factors by offering discounted rates (\$/kWh).



EXAMPLE 7.10 Power Factor Correction

Problem

A capacitor is used to correct the power factor of the 100 kW and lagging $\text{pf} = 0.7$ load of Figure 7.23. Determine the reactive power of the load alone, and compute the capacitance required for a corrected power factor $\text{pf} = 1$.

Solution

Known Quantities: Source voltage; load power and power factor.

Find:

1. The reactive power Q of the load alone.
2. The capacitance C required for a corrected power factor $\text{pf} = 1$.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_s = 480\angle 0 \text{ V rms}$; $P = 10^5 \text{ W}$; $\text{pf} = 0.7$ lagging for the load; $\omega = 377 \text{ rad/s}$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise.

Analysis:

1. For the load alone, $\text{pf} = 0.7$ lagging or $\cos(\theta_Z) = 7/10$, and the power triangle has the shape shown in Figure 7.24. The real power is given as $P = 100 \text{ kW}$, so the reactive power of the load can be computed using the relative triangle dimensions to be:

$$Q = P \tan(\theta_Z) = (100 \text{ kW})(\sqrt{51}/7) = 102 \text{ kVAR}$$

Since the power factor is lagging, the reactive power is positive as indicated in Table 7.2 and shown in the power triangle of Figure 7.25.

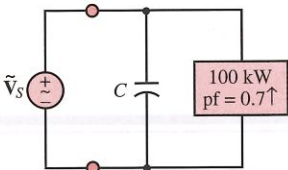


Figure 7.23

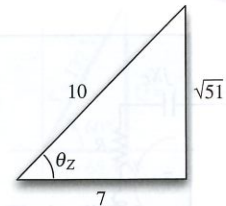


Figure 7.24 Relative dimensions of power triangle

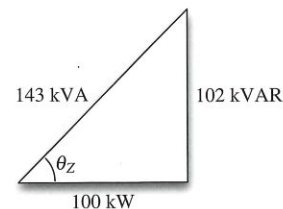


Figure 7.25 Power triangle

2. To set the corrected power factor to $\text{pf} = 1$ the capacitance must contribute -102 kVAR of reactive power. That is:

$$Q_C = \Delta Q = Q_{\text{final}} - Q_{\text{initial}} = 0 - 102 \text{ kVAR} = -102 \text{ kVAR}$$

Since the voltage across capacitor $\tilde{V}_C$ equals the source voltage $\tilde{V}_s$, the reactive power of the capacitor is:

$$Q_C = \frac{|\tilde{V}_C|^2}{|X_C|} \sin(-90^\circ) = -(\omega C)|\tilde{V}_s|^2 = -(377)(480^2)C$$

Thus, to correct the power factor to $\text{pf} = 1$ (zero total reactive power), the capacitor must satisfy:

$$Q_C = -(377)(480^2)C = -102 \text{ kVAR}$$

or

$$C = \frac{102 \text{ kVAR}}{(377)(480^2)} = 1,175 \mu\text{F}$$

Use trigonometry and/or the Pythagorean theorem to show that the apparent power $|S| = 143 \text{ kVA}$, as indicated in Figure 7.25.

Comments: Note that it is not necessary to know the load impedance to perform power factor correction; however, it is a useful exercise to compute the equivalent impedance seen by $\tilde{V}_s$ and check that $\cos(\theta_Z) = 0.7$.



EXAMPLE 7.11 Power Factor Correction

Problem

Figure 7.26 shows a second load added to the circuit of Figure 7.23. Determine the capacitance required for an overall corrected power factor $\text{pf} = 1$. Draw the phasor diagram showing the relationship between $\tilde{I}_C$, $\tilde{I}_1$, and $\tilde{I}_2$.

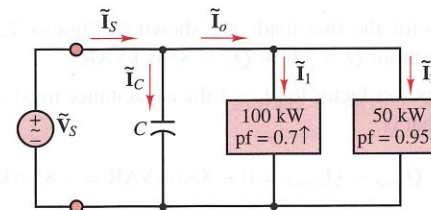


Figure 7.26

Solution

Known Quantities: Source voltage; load power and power factor.

Find:

1. The total reactive power of loads 1 and 2.
2. The capacitance C required for an overall power factor $\text{pf} = 1$.
3. $\tilde{I}_C$, $\tilde{I}_1$, and $\tilde{I}_2$, and construct a phasor diagram of these currents.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_s = 480\angle 0^\circ \text{ V rms}$; $P_1 = 100 \text{ kW}$; $\text{pf}_1 = 0.7$ lagging; $P_2 = 50 \text{ kW}$; $\text{pf}_2 = 0.95$ leading; $\omega = 377 \text{ rad/s}$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise.

Analysis:

1. Compute $\tilde{I}_1$ and $\tilde{I}_2$ using the relation $P = |\tilde{V}||\tilde{I}|\text{pf}$.

$$P_1 = |\tilde{V}_s||\tilde{I}_1|\cos(\theta_1) \rightarrow |\tilde{I}_1| = \frac{P_1}{|\tilde{V}_s|\cos(\theta_1)} \approx 298 \text{ A}$$

and

$$\angle \tilde{V}_s = \angle \tilde{I}_1 + \theta_{Z_1} \rightarrow \angle \tilde{I}_1 = \angle \tilde{V}_s - \theta_{Z_1} = 0 - \cos^{-1}(0.7) \approx -0.795 \text{ rad}$$

It is important to keep in mind that although inverse trigonometric functions are double-valued [e.g., $\cos^{-1}(0.7) \approx \pm 0.795 \text{ rad}$], the power factor for load 1 is lagging such that $\theta_{Z_1} = +0.795 \text{ rad}$ is the correct choice.

Similarly, for load 2:

$$P_2 = |\tilde{V}_s||\tilde{I}_2|\cos(\theta_2) \rightarrow |\tilde{I}_2| = \frac{P_2}{|\tilde{V}_s|\cos(\theta_2)} \approx 110 \text{ A}$$

and

$$\angle \tilde{V}_s = \angle \tilde{I}_2 + \theta_{Z_2} \rightarrow \angle \tilde{I}_2 = \angle \tilde{V}_s - \theta_{Z_2} = 0 - \cos^{-1}(0.95) \approx +0.318 \text{ rad}$$

The power factor for load 2 is leading such that $\theta_{Z_2} = -0.318 \text{ rad}$ is the correct choice.

Now use the given data and the relation $Q = P \tan(\theta_Z)$ to compute the reactive power for each load.

$$Q_1 = P_1 \tan(+0.795 \text{ rad}) \approx +102 \text{ kVAR}$$

and

$$Q_2 = P_2 \tan(-0.318 \text{ rad}) \approx -16.4 \text{ kVAR}$$

The power triangles for the two loads are shown in Figures 7.27 and 7.28. The total reactive power is therefore $Q = Q_1 + Q_2 \approx 85.6 \text{ kVAR}$.

2. To set the corrected power factor to $\text{pf} = 1$ the capacitance must contribute -85.6 kVAR of reactive power. That is:

$$Q_C = \Delta Q = Q_{\text{final}} - Q_{\text{initial}} = 0 - 85.6 \text{ kVAR} = -85.6 \text{ kVAR}$$

For a capacitor alone its reactive power is:

$$Q_C = \frac{|\tilde{V}_C|^2}{X_C} = -(\omega C)|\tilde{V}_s|^2 = -(377)(480^2)C$$

Thus, to correct the power factor to $\text{pf} = 1$ (zero total reactive power), the capacitor must satisfy:

$$Q_C = -(377)(480^2)C = -85.6 \text{ kVAR}$$

or

$$C = \frac{85.6 \text{ kVAR}}{(377)(480^2)} \approx 985 \mu\text{F}$$

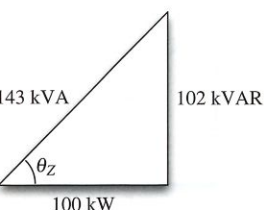


Figure 7.27 Power triangle for load 1.

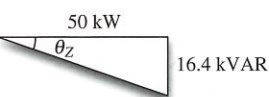


Figure 7.28 Power triangle for load 2.

3. To compute the capacitor current it is not possible to use $P = |\tilde{V}||\tilde{I}|\text{pf}$ because $P = 0$ and $\text{pf} = 0$ for a capacitor. Instead, the generalized Ohm's law provides an alternative approach.

$$\tilde{V}_C = \tilde{I}_C Z_C \rightarrow |\tilde{I}_C| = \frac{|\tilde{V}_C|}{|Z_C|} = \omega C |\tilde{V}_C| \approx 178.2 \text{ A}$$

where $\tilde{V}_C = \tilde{V}_s$. The phase angle of $\tilde{I}_C$ is:

$$\angle \tilde{I}_C = \angle \tilde{V}_C - \theta_{Z_C} = 0 - (-\pi/2) = +\pi/2 \text{ rad}$$

The current phasor diagram can now be drawn as shown in Figure 7.29.

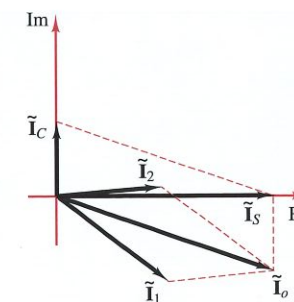


Figure 7.29

Comment: The power triangle suggests that the capacitor current can also be calculated using the relation $Q_C = |\tilde{V}_C||\tilde{I}_C|\sin(\theta_C)$, where $\theta_C = -\pi/2$ and $Q_C = |\tilde{V}_C|^2/X_C = -(\omega C)|\tilde{V}_C|^2$. Try it!

CHECK YOUR UNDERSTANDING

Compute the magnitude of $\tilde{I}_s$ after the power factor correction in Example 7.8.

Answer: 0.585 A

CHECK YOUR UNDERSTANDING

Two cases of the voltage across and the current through a load are given below. Determine the power factor of the load, and whether it is leading or lagging, for each case.

- a. $v(t) = 540 \cos(\omega t + 15^\circ) \text{ V}$, $i(t) = 2 \cos(\omega t + 47^\circ) \text{ A}$
- b. $v(t) = 155 \cos(\omega t - 15^\circ) \text{ V}$, $i(t) = 2 \cos(\omega t - 22^\circ) \text{ A}$

Answer: a. 0.848, leading; b. 0.9925, lagging

CHECK YOUR UNDERSTANDING

Determine if a load is capacitive or inductive, given the following facts:

- a. $\text{pf} = 0.87$, leading
- b. $\text{pf} = 0.42$, leading
- c. $v(t) = 42 \cos(\omega t) \text{ V}$, $i(t) = 4.2 \sin(\omega t) \text{ A}$ [Hint: $\sin(\omega t)$ lags $\cos(\omega t)$.]
- d. $v(t) = 10.4 \cos(\omega t - 22^\circ) \text{ V}$, $i(t) = 0.4 \cos(\omega t - 22^\circ) \text{ A}$

Answer: a. Capacitive; b. Capacitive; c. inductive; d. neither (resistive)

CHECK YOUR UNDERSTANDING

Compute the power factor for an inductive load with $L = 100 \text{ mH}$ in series with $R = 0.4 \Omega$. Assume $\omega = 377 \text{ rad/s}$.

Answer: pf = 0.106, lagging

The Wattmeter

The instrument used to measure power is called a wattmeter. The external part of a wattmeter consists of four connections and a metering mechanism that displays the amount of real power dissipated by a circuit. The external and internal appearance of a wattmeter is depicted in Figure 7.30. Inside the wattmeter are two coils: a current-sensing coil and a voltage-sensing coil. In this example, we assume for simplicity that the impedance of the current-sensing coil Z_I is zero and that the impedance of the voltage-sensing coil Z_V is infinite. In practice, this will not necessarily be true; some correction mechanism will be required to account for the impedance of the sensing coils.

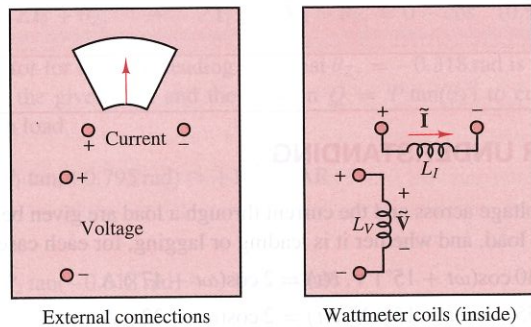


Figure 7.30

A wattmeter should be connected as shown in Figure 7.31 to provide both current and voltage measurements. We see that the current-sensing coil is placed in series with the

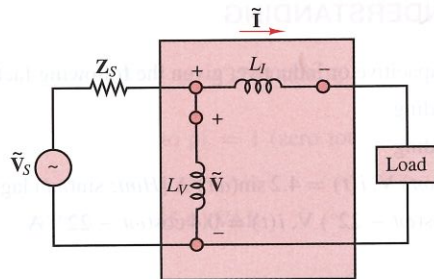


Figure 7.31

(Continued)

load and that the voltage-sensing coil is placed in parallel with the load. In this manner, the wattmeter is seeing the current through and the voltage across the load. Remember that the power dissipated by a circuit element is related to these two quantities. The wattmeter, then, is constructed to provide a readout of the real power absorbed by the load: $P = \text{Re}(S) = \text{Re}(\tilde{V}\tilde{I}^*)$.

Problem:

- For the circuit shown in Figure 7.32, show the connections of a wattmeter between the ideal voltage source and the load and find the power dissipated by the load.
- Show the connections that will determine the power dissipated by R_2 . What should the meter read?

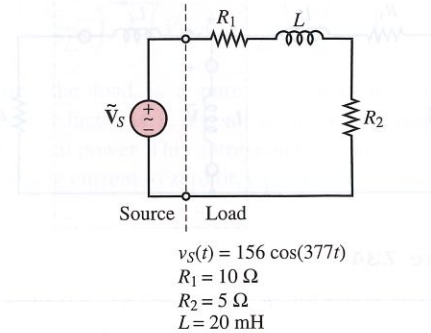


Figure 7.32

Solution:

- To measure the power dissipated by the load, we must know the current through and the voltage across the entire load circuit. This means that the wattmeter must be connected as shown in Figure 7.33. The wattmeter should read

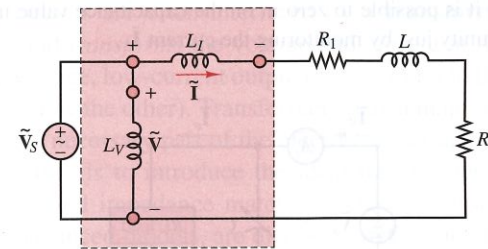


Figure 7.33

$$\begin{aligned}
 P &= \text{Re}(\tilde{V}_s \tilde{I}^*) = \text{Re} \left[\left(\frac{156}{\sqrt{2}} \angle 0^\circ \right) \left(\frac{(156/\sqrt{2}) \angle 0}{R_1 + R_2 + j\omega L} \right)^* \right] \\
 &= \text{Re} \left[110 \angle 0^\circ \left(\frac{110 \angle 0}{15 + j7.54} \right)^* \right] \\
 &= \text{Re} \left[110 \angle 0^\circ \left(\frac{110 \angle 0}{16.79 \angle 0.466} \right)^* \right] = \text{Re} \frac{110^2}{16.79 \angle (-0.466)} \\
 &= \text{Re} (720.67 \angle 0.466) \\
 &= 643.88 \text{ W}
 \end{aligned}$$

(Continued)

(Concluded)

2. To measure the power dissipated by R_2 alone, we must measure the current through R_2 and the voltage across R_2 alone. The connection is shown in Figure 7.34. The meter will read

$$P = |\tilde{I}|^2 R_2 = \left[\frac{110}{(15^2 + 7.54^2)^{1/2}} \right]^2 \times 5 = \frac{110^2}{15^2 + 7.54^2} \times 5 = 215 \text{ W}$$

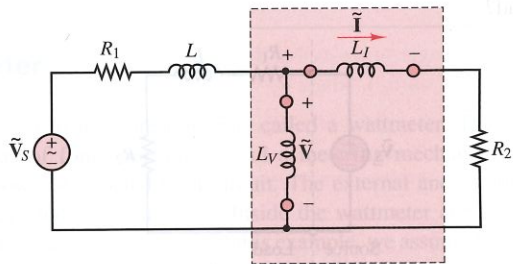


Figure 7.34

FOCUS ON MEASUREMENTS

Power Factor

Problem:

A capacitor is being used to correct the power factor of a load to unity, as shown in Figure 7.35. The capacitor value is varied, and measurements of the total current are taken. Explain how it is possible to zero in on the capacitance value necessary to bring the power factor to unity just by monitoring the current $\tilde{I}_s$.

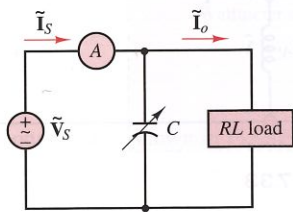


Figure 7.35

Solution:

The current through the load is

$$\begin{aligned} \tilde{I}_o &= \frac{\tilde{V}_s \angle 0^\circ}{R + j\omega L} = \frac{\tilde{V}_s}{R^2 + \omega^2 L^2} (R - j\omega L) \\ &= \frac{\tilde{V}_s R}{R^2 + \omega^2 L^2} - j \frac{\tilde{V}_s \omega L}{R^2 + \omega^2 L^2} \end{aligned}$$

(Continued)

(Concluded)

The current through the capacitor is

$$\tilde{I}_c = \frac{\tilde{V}_s \angle 0^\circ}{1/j\omega C} = j\tilde{V}_s \omega C$$

The source current to be measured is

$$\tilde{I}_s = \tilde{I}_o + \tilde{I}_c = \frac{\tilde{V}_s R}{R^2 + \omega^2 L^2} + j \left(\tilde{V}_s \omega C - \frac{\tilde{V}_s \omega L}{R^2 + \omega^2 L^2} \right)$$

The magnitude of the source current is

$$|\tilde{I}_s| = \sqrt{\left(\frac{\tilde{V}_s R}{R^2 + \omega^2 L^2} \right)^2 + \left(\tilde{V}_s \omega C - \frac{\tilde{V}_s \omega L}{R^2 + \omega^2 L^2} \right)^2}$$

We know that when the load is a pure resistance, then the current and voltage are in phase, the power factor is 1, and all the power delivered by the source is dissipated by the load as real power. This corresponds to equating the imaginary part of the expression for the source current to zero or, equivalently, to the following expression:

$$\frac{\tilde{V}_s \omega L}{R^2 + \omega^2 L^2} = \tilde{V}_s \omega C$$

in the expression for $|\tilde{I}_s|$. Thus, the magnitude of the source current is actually a minimum when the power factor is unity! It is therefore possible to “tune” a load to a unity pf by observing the readout of the ammeter while changing the value of the capacitor and selecting the capacitor value that corresponds to the lowest source current value.

7.4 TRANSFORMERS

Two separate AC circuits are often interfaced by a **transformer**, which acts as a magnetic coupling and *transforms* the voltage and current at the interface (e.g., by matching the high-voltage, low-current output of one circuit to the low-voltage, high-current input required by the other). Transformers play a major role in electric power engineering and are a necessary part of the electric power distribution network. The objective of this section is to introduce the ideal transformer and the concepts of impedance reflection and impedance matching. The operations of practical transformers, and more advanced models, are discussed in Chapter 18.

The Ideal Transformer

The ideal transformer consists of two coils coupled to each other by a magnetic medium. There is no conducting electrical connection between the coils. The input side of a transformer is known as the **primary** while the output side is known as the **secondary**. The number of turns in the primary and secondary coils are designated n_1 and n_2 , respectively. The **turns ratio** N is defined by:

$$N = \frac{n_2}{n_1} \quad (7.29)$$

Figure 7.36 illustrates the convention by which voltages and currents are usually assigned at a transformer. The solid black dots in Figure 7.36 are used to mark coil terminals that have the same polarity.

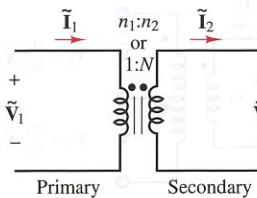


Figure 7.36 Ideal transformer

Recall from Faraday's law that each coil experiences *self-induction* in that a time-varying current through a coil produces a time-varying magnetic flux through the coil itself, which, in turn, induces a potential difference opposing the time-varying magnetic flux. The net effect of this self-induction is expressed by the inductance L of a coil. However, when two coils are present, as in a transformer, both coils also experience *mutual induction* in that some of the time-varying magnetic flux due to one coil passes through the other coil and induces another opposing potential difference. The net effect of the mutual induction is expressed by the mutual inductance M of the two coils. Both L and M contribute to the behavior of a transformer.

Notice the emphasis on *time variations* in the previous paragraph. One result of Faraday's law is that a leave in current through a coil, which generates a constant magnetic field, induces no opposing reaction within the coil itself (no self-induction) nor within any nearby coil (no mutual induction). Instead, a coil acts as a short-circuit in the presence of leave in current and transformers perform no useful function in DC circuits. See Chapter 18 for further discussion of Faraday's law as it relates to electromechanics.

As depicted in Figure 7.36, the relationships between primary and secondary currents and voltages in an ideal transformer are:

$$\begin{aligned}\tilde{V}_2 &= N\tilde{V}_1 \\ \tilde{I}_2 &= \frac{\tilde{I}_1}{N}\end{aligned}$$

Ideal transformer (7.30)

When $N > 1$, $|\tilde{V}_2| > |\tilde{V}_1|$ and a transformer is called a **step-up transformer**. When $N < 1$, $|\tilde{V}_2| < |\tilde{V}_1|$ and a transformer is called a **step-down transformer**. Either side of an ideal transformer can be used as the primary; thus, to produce a step-up transformer from a step-down transformer one only need exchange the primary and secondary connections. (Exchanging the primary and secondary by mistake can lead to significant dangers in a laboratory experiment!) Finally, when $N = 1$, a transformer is called an **isolation transformer**, which can be used to electrically couple or isolate two circuits and adjust the output and input impedances at the interface of two circuits.

A comparison of the complex power at the primary and secondary terminals of an ideal transformer reveals that they are the same:

$$S_1 = \tilde{I}_1^* \tilde{V}_1 = N \tilde{I}_2^* \frac{\tilde{V}_2}{N} = \tilde{I}_2^* \tilde{V}_2 = S_2 \quad (7.31)$$

That is, **ideal transformers conserve power**.

As shown in Figure 7.37, the secondary coil of many practical transformers is *center-tapped*, which splits the secondary voltage into two equal halves. This type of transformer is found at the entry of a power line into a household, where a high-voltage primary is transformed to 240 V as well as split into two 120-V lines. Referring to Figure 7.37, $\tilde{V}_2$ and $\tilde{V}_3$ would both provide 120 V for common household appliances while $(\tilde{V}_2 + \tilde{V}_3)$ would provide 240 V for higher-powered devices, such as clothes dryers and electric ranges.

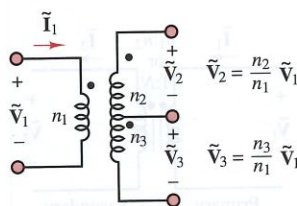


Figure 7.37 Center-tapped transformer

Impedance Reflection

Transformers are commonly used to couple one AC circuit to another, as depicted in Figure 7.38, where an AC Thévenin source network is connected to a load Z_2 by means of a transformer.

The equivalent impedance seen by the Thévenin source is that of the entire network to the right of terminals a and b . Applying the definition of equivalent impedance and using the ideal transformer relations from equation 7.30, the result is:

$$\begin{aligned}Z_1 &\equiv \frac{\tilde{V}_1}{\tilde{I}_1} = \frac{\tilde{V}_2}{N} \frac{1}{N\tilde{I}_2} \\ &= \frac{1}{N^2} \frac{\tilde{V}_2}{\tilde{I}_2} \\ &= \frac{1}{N^2} Z_2\end{aligned} \quad (7.32)$$

Thus, the equivalent impedance seen by the AC Thévenin source is the load impedance Z_2 reduced by the factor $1/N^2$.

Likewise, the equivalent network seen by Z_2 is the Thévenin equivalent of the entire network to the left of terminals c and d . When Z_2 is replaced by an open-circuit, $\tilde{I}_2 = 0$ and the Thévenin (open-circuit) voltage is:

$$\tilde{V}_T = (\tilde{V}_2)_{OC} = N\tilde{V}_1 \quad (7.33)$$

However, since $\tilde{I}_1 = N\tilde{I}_2 = 0$, the voltage drop across Z_S is zero such that $\tilde{V}_1 = \tilde{V}_S$ with the result:

$$\tilde{V}_T = (\tilde{V}_2)_{OC} = N\tilde{V}_S \quad (7.34)$$

When Z_2 is replaced by a short-circuit, $\tilde{V}_2 = 0$ and the short-circuit current is:

$$(\tilde{I}_2)_{SC} = \frac{\tilde{I}_1}{N} \quad (7.35)$$

However, since $\tilde{V}_1 = \tilde{V}_2/N = 0$, the voltage drop across Z_S is $\tilde{V}_S$ such that $\tilde{I}_1 = \tilde{V}_S/Z_S$ with the result:

$$(\tilde{I}_2)_{SC} = \frac{1}{N} \frac{\tilde{V}_S}{Z_S} \quad (7.36)$$

Thus, the Thévenin equivalent impedance seen by Z_2 is:

$$Z_T = \frac{(\tilde{V}_2)_{OC}}{(\tilde{I}_2)_{SC}} = N\tilde{V}_S \frac{NZ_S}{\tilde{V}_S} = N^2 Z_S \quad (7.37)$$

Thus, the equivalent impedance seen by Z_2 is the source impedance Z_S multiplied by N^2 .

Figure 7.39 summarizes and illustrates these effects, which are known as **impedance reflection** across a transformer and which play an important role in power transfer.

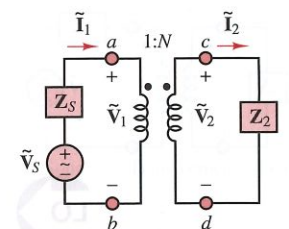


Figure 7.38 Operation of an ideal transformer

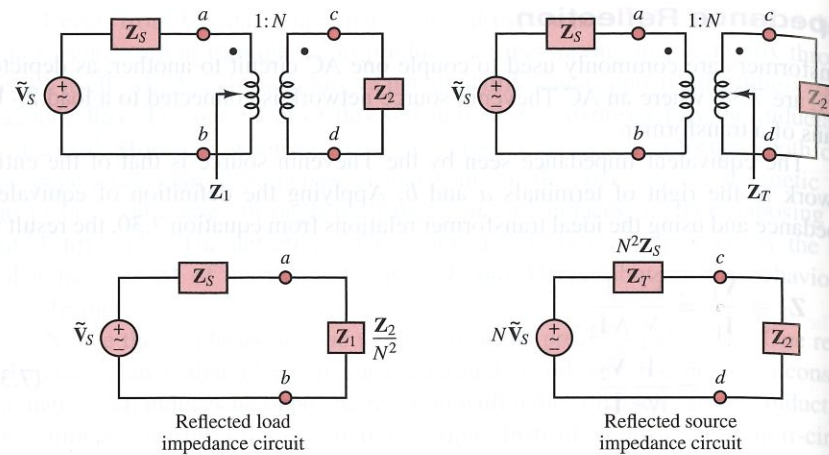


Figure 7.39 Impedance reflection across a transformer

Maximum Power Transfer

Recall that in resistive DC circuits, maximum power is transferred to a load when the load equals the Thévenin equivalent resistance of the source network. For AC circuits, the analogous maximum power transfer condition is known as **impedance matching**.

Consider the general form of an AC circuit, shown in Figure 7.40, and assume that the source impedance Z_T is:

$$Z_T = R_T + jX_T \quad (7.38)$$

What value of the load Z_o results in the maximum *real power* transfer to the load itself? The real power absorbed by the load is:

$$P_o = \tilde{V}_o \tilde{I}_o \cos \theta_{Z_o} = \text{Re}(\tilde{V}_o \tilde{I}_o^*) \quad (7.39)$$

Apply voltage division and the generalized Ohm's law to find:

$$\tilde{V}_o = \frac{Z_o}{Z_T + Z_o} \tilde{V}_T \quad \tilde{I}_o = \frac{\tilde{V}_T}{Z_T + Z_o} \quad (7.40)$$

Let $Z_o = R_o + jX_o = |Z_o| \cos \theta_{Z_o} + j|Z_o| \sin \theta_{Z_o}$, and since $\tilde{V}_o = |\tilde{V}_o|$ and $\tilde{I}_o = |\tilde{I}_o|$, the real power absorbed by the load can be expressed as:

$$P_o = \frac{|Z_o|}{|Z_T + Z_o|} \tilde{V}_T \times \frac{1}{|Z_T + Z_o|} \tilde{V}_T \times \frac{R_o}{|Z_o|} \quad (7.41)$$

Or, after simplification:

$$P_o = \frac{R_o}{|Z_T + Z_o|^2} \tilde{V}_T^2 = \frac{R_o}{(R_T + R_o)^2 + (X_T + X_o)^2} \tilde{V}_T^2 \quad (7.42)$$

The condition for the maximum value of P_o can be found by solving:

$$\frac{\partial P_o}{\partial R_o} dR_o + \frac{\partial P_o}{\partial X_o} dX_o = 0 \quad (7.43)$$

or

$$\frac{\partial P_o}{\partial R_o} = 0 \quad \text{and} \quad \frac{\partial P_o}{\partial X_o} = 0 \quad (7.44)$$

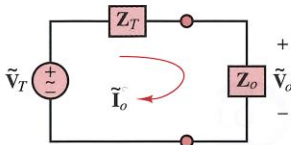


Figure 7.40 The maximum power transfer problem in AC circuits

Both of these conditions are satisfied when $R_o = R_T$ and $X_o = X_T$. That is, the condition for maximum real power transfer to a load is $Z_o = Z_T^*$.

$$Z_o = Z_T^* \quad \text{Maximum power transfer} \quad (7.45)$$

Maximum power is transferred to the load when its impedance equals the complex conjugate of the Thévenin equivalent impedance of the source. When this condition is satisfied, the load and source impedances are *matched*.

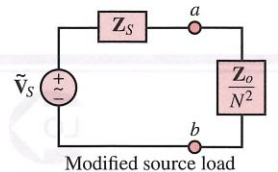
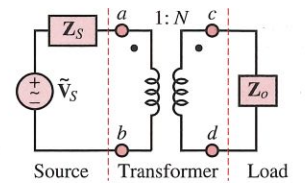


Figure 7.41 Maximum power transfer in an AC circuit with a transformer

In some cases, it may not be possible to match the load to the source because of practical limitations. In these situations, it may be possible to use a transformer as the interface between the source and the load to achieve maximum power transfer. Figure 7.41 illustrates how the reflected load impedance, as seen by the source, is equal to Z_o/N^2 , such that the condition for maximum power transfer is:

$$\begin{aligned} \frac{Z_o}{N^2} &= Z_S^* \\ R_o &= N^2 R_S \\ X_o &= -N^2 X_S \end{aligned} \quad (7.46)$$

EXAMPLE 7.12 Ideal Transformer Turns Ratio

Problem

We require a transformer (see Figure 7.42) to output 500 mA at 24 V from a 120 V rms input line source. The primary has $n_1 = 3,000$ turns. How many turns are required in the secondary? What is the primary current?

Solution

Known Quantities: Primary and secondary voltages; secondary current; number of turns in the primary coil.

Find: n_2 and $\tilde{I}_1$.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_1 = 120$ V; $\tilde{V}_2 = 24$ V; $\tilde{I}_2 = 500$ mA; $n_1 = 3,000$ turns.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians.

Analysis: Use equation 7.30 to compute the number of turns in the secondary coil:

$$\frac{\tilde{V}_1}{n_1} = \frac{\tilde{V}_2}{n_2} \quad n_2 = n_1 \frac{\tilde{V}_2}{\tilde{V}_1} = 3,000 \times \frac{24}{120} = 600 \text{ turns}$$

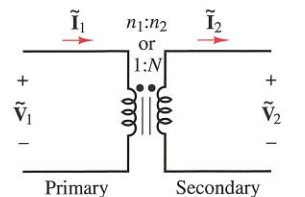


Figure 7.42 Example 7.12

Again, use equations 7.29 and 7.30 to compute the primary current:

$$n_1 \tilde{I}_1 = n_2 \tilde{I}_2 \quad \tilde{I}_1 = \frac{n_2}{n_1} \tilde{I}_2 = \frac{600}{3,000} \times 500 = 100 \text{ mA}$$

Comment: Note that since the transformer does not affect the phase of the voltages and currents, it was possible to solve the problem using only the rms amplitudes.



EXAMPLE 7.13 Center-Tapped Transformer

Problem

An ideal center-tapped power transformer (Figure 7.43) has a 4,800-V primary and a 240-V secondary. The center-tap is located such that $\tilde{V}_2 = \tilde{V}_3 = 120 \text{ V}$. Three resistive loads are attached to the secondary terminals. Compute the current in the primary assuming that R_2 , R_3 , and R_4 each absorb P_2 , P_3 , and P_4 , respectively. Also compute the current through each load and the resistance of each load.

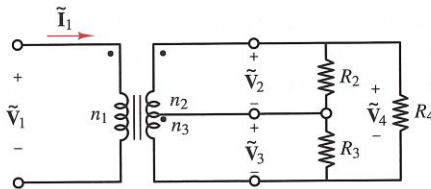


Figure 7.43 Example 7.13

Solution

Known Quantities: Primary and secondary voltages; load power ratings.

Find $\tilde{I}_{\text{primary}} = |\tilde{I}|$

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_1 = 4,800 \text{ V rms}$; $\tilde{V}_2 = 120 \text{ V rms}$; $\tilde{V}_3 = 120 \text{ V rms}$; $P_2 = 5,000 \text{ W}$; $P_3 = 1,000 \text{ W}$; $P_4 = 1,500 \text{ W}$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise. The transformer is ideal.

Analysis: Power is conserved for an ideal transformer; thus:

$$S_{\text{primary}} = S_{\text{secondary}}$$

Since each load is purely resistive, $\theta_Z = 0$ and $\text{pf} = \cos \theta_Z = 1$ such that:

$$|S|_{\text{secondary}} = P_{\text{secondary}} = P_2 + P_3 + P_4 = 7,500 \text{ W}$$

Since $|S|_{\text{primary}} = |S|_{\text{secondary}}$:

$$\tilde{V}_{\text{primary}} \times \tilde{I}_{\text{primary}} = 7,500 \text{ W}$$

Thus:

$$\tilde{I}_{\text{primary}} = \frac{7,500 \text{ W}}{4,800 \text{ V rms}} = 1.5625 \text{ A rms}$$

The current through each resistor is simply:

$$\tilde{I}_2 = \frac{P_2}{\tilde{V}_2} = \frac{5,000 \text{ W}}{120 \text{ V rms}} \approx 41.7 \text{ A rms}$$

$$\tilde{I}_3 = \frac{P_3}{\tilde{V}_3} = \frac{1,000 \text{ W}}{120 \text{ V rms}} \approx 8.3 \text{ A rms}$$

$$\tilde{I}_4 = \frac{P_4}{\tilde{V}_2 + \tilde{V}_3} = \frac{1,500 \text{ W}}{240 \text{ V rms}} = 6.25 \text{ A rms}$$

The resistor values are:

$$\tilde{R}_2 = \frac{P_2}{\tilde{I}_2^2} = 2.88 \Omega$$

$$\tilde{R}_3 = \frac{P_3}{\tilde{I}_3^2} = 14.4 \Omega$$

$$\tilde{R}_4 = \frac{P_4}{\tilde{I}_4^2} = 38.4 \Omega$$

Comments: The calculations in this example were particularly straightforward because the load was purely resistive, such that $\theta_Z = 0$, the power triangle is flat, and the apparent power S equals the real power P . When the load is complex, $\theta_Z > 0$, the power triangle is not flat, and the apparent power S equals $P \cos \theta_Z$. Then, the calculations are more complicated.

Also, KCL can be used to determine the current drawn from/to the outside- and center-taps. Try it!

EXAMPLE 7.14 Use of Transformers to Improve Power Line Efficiency



Problem

Figure 7.44 illustrates the use of transformers in electric power transmission lines. The line voltage is transformed before and after being transmitted over long distances. This example illustrates the efficiency gained through the use of transformers. For the sake of simplicity, ideal transformers and simple resistive models for the generator, transmission line, and load have been assumed.

Solution

Known Quantities: Values of circuit elements.

Find: Calculate the power transfer efficiency for the two circuits of Figure 7.44.

Schematics, Diagrams, Circuits, and Given Data: Step-up transformer turns ratio is N , step-down transformer turns ratio is $M = 1/N$. All transformers are ideal.

Assumptions: None.

Analysis: Since the load and source currents are equal in Figure 7.44(a), the power transmission efficiency is:

$$\eta = \frac{P_{\text{load}}}{P_{\text{source}}} = \frac{\tilde{V}_{\text{load}} \tilde{I}_{\text{load}}}{\tilde{V}_{\text{source}} \tilde{I}_{\text{load}}} = \frac{\tilde{V}_{\text{load}}}{\tilde{V}_{\text{source}}} = \frac{R_{\text{load}}}{R_{\text{source}} + R_{\text{line}} + R_{\text{load}}}$$

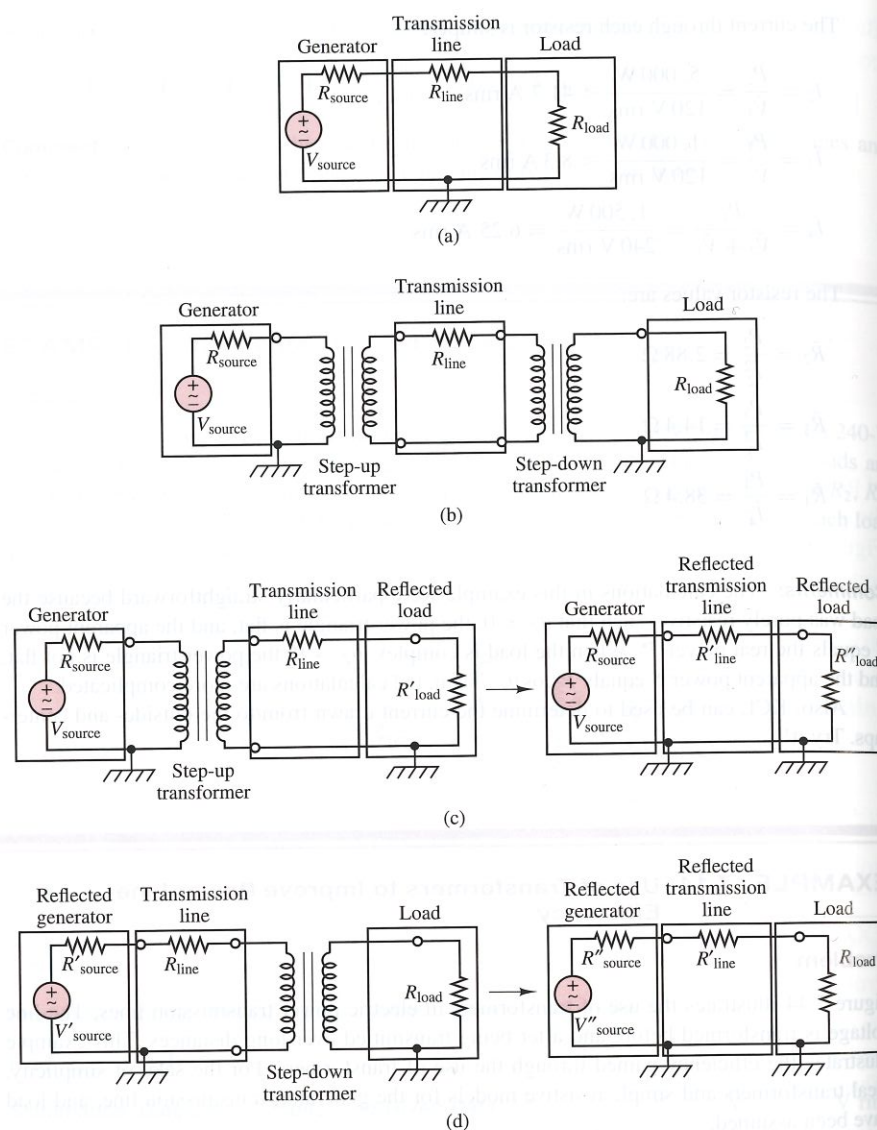


Figure 7.44 Electric power transmission: (a) direct power transmission; (b) power transmission with transformers; (c) equivalent circuit seen by generator; (d) equivalent circuit seen by load.

In Figure 7.44(b), transformers are introduced between each of the three portions of the overall circuit. The equivalent load resistance seen by the transmission line (or “reflected” by the step-down transformer) is found from equation 7.32 to be:

$$R'_{\text{load}} = \frac{1}{M^2} R_{\text{load}} = N^2 R_{\text{load}}$$

Now, the step-up transformer sees the equivalent impedance $R'_{\text{load}} + R_{\text{line}}$. The resistance seen by the generator (or “reflected” by the step-up transformer) is:

$$R''_{\text{load}} = \frac{1}{N^2} (R'_{\text{load}} + R_{\text{line}}) = R_{\text{load}} + \frac{1}{N^2} R_{\text{line}}$$

These transformations are depicted in Figure 7.44(c). The effect of the two transformers is to reduce the line resistance seen by the source by N^2 . The source current is:

$$\tilde{I}_{\text{source}} = \frac{\tilde{V}_{\text{source}}}{R_{\text{source}} + R''_{\text{load}}} = \frac{\tilde{V}_{\text{source}}}{R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}}$$

The source power is:

$$P_{\text{source}} = \frac{\tilde{V}_{\text{source}}^2}{R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}}$$

The same process can be repeated starting from the left and reflecting the source circuit to the right of the step-up transformer:

$$\tilde{V}'_{\text{source}} = N \tilde{V}_{\text{source}} \quad \text{and} \quad R'_{\text{source}} = N^2 R_{\text{source}}$$

Now the circuit to the left of the step-down transformer comprises the series combination of $\tilde{V}'_{\text{source}}$, R'_{source} , and R_{line} , which can be reflected to the right of the step-down transformer to obtain $\tilde{V}''_{\text{source}} = M \tilde{V}'_{\text{source}} = \tilde{V}_{\text{source}}$, $R''_{\text{source}} = M^2 R'_{\text{source}} = R_{\text{source}}$, $R''_{\text{line}} = M^2 R_{\text{line}}$, and R_{load} in series. These transformations are depicted in Figure 7.44(d). Thus, the load voltage, current, and power are:

$$\tilde{I}_{\text{load}} = \frac{\tilde{V}_{\text{source}}}{R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}}$$

$$\tilde{V}_{\text{load}} = \tilde{V}_{\text{source}} \frac{R_{\text{load}}}{R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}}$$

$$P_{\text{load}} = \tilde{I}_{\text{load}} \tilde{V}_{\text{load}} = \frac{\tilde{V}_{\text{source}}^2 R_{\text{load}}}{[R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}]^2}$$

Finally, the power efficiency can be computed as the ratio of the load to source power:

$$\eta = \frac{P_{\text{load}}}{P_{\text{source}}} = \frac{\tilde{V}_{\text{source}}^2 R_{\text{load}}}{[R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}]^2} \frac{R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}}{\tilde{V}_{\text{source}}^2}$$

$$= \frac{R_{\text{load}}}{R_{\text{source}} + (1/N^2)R_{\text{line}} + R_{\text{load}}}$$

Notice that the power transmission efficiency calculated for Figure 7.44(a) was improved by reducing the effect of the line resistance by a factor of $1/N^2$.

EXAMPLE 7.15 Maximum Power Transfer Through a Transformer

Problem

Find the transformer turns ratio N and the load reactance X_o that results in maximum power transfer in the transformer shown in Figure 7.45.

Solution

Known Quantities: Source voltage, frequency, and impedance; load resistance.

Find: Transformer turns ratio and load reactance.

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_S = 240 \angle 0^\circ \text{ V rms}$; $R_S = 10 \Omega$; $L_S = 0.1 \text{ H}$; $R_o = 400 \Omega$; $\omega = 377 \text{ rad/s}$.

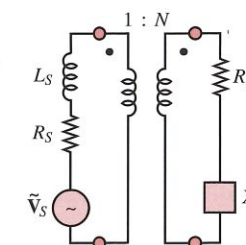


Figure 7.45

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians. The transformer is ideal.

Analysis: The requirements for maximum power transfer, as given by equation 7.46, are $R_o = N^2 R_s$ and $X_o = -N^2 X_s = -N^2(\omega \times 0.1)$. Thus:

$$N^2 = \frac{R_o}{R_s} = \frac{400}{10} = 40 \quad N = \sqrt{40} = 6.325$$
$$X_o = -40 \times 37.7 = -1,508 \, \Omega$$

Thus, the load reactance should be a capacitor with value:

$$C = -\frac{1}{X_o \omega} = -\frac{1}{(-1,508)(377)} = 1.76 \, \mu\text{F}$$

CHECK YOUR UNDERSTANDING

With reference to Example 7.12, compute the number of primary turns required if $n_2 = 600$ but the transformer is required to deliver 1 A. What is the primary current now?

Answers: $n_1 = 3,000; I_1 = 200 \text{ mA}$

CHECK YOUR UNDERSTANDING

If the transformer of Example 7.13 has 300 turns in the secondary coil, how many turns will the primary require?

Answer: $n_2 = 6,000$

CHECK YOUR UNDERSTANDING

Assume that the generator produces a source voltage of 480 V rms, and that $N = 300$. Further assume that the source impedance is $2 \, \Omega$, the line impedance is also $2 \, \Omega$, and that the load impedance is $8 \, \Omega$. Calculate the efficiency improvement for the circuit of Figure 7.37(b) over the circuit of Figure 7.37(a).

Answer: 80% vs. 67%

CHECK YOUR UNDERSTANDING

The transformer shown in Figure 7.46 is ideal. Assume that $Z_s = 1,800 \, \Omega$ and $Z_o = 8 \, \Omega$ to find the turns ratio N that will ensure maximum power transfer to the load.

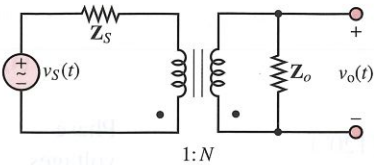


Figure 7.46

Now assume that $N = 5.4$ and $Z_o = 2 + j10 \, \Omega$. Find the source impedance Z_s that will ensure maximum power transfer to the load.

Answers: $N = 0.0667; Z_s = 0.0686 - j0.3429 \, \Omega$

7.5 THREE-PHASE POWER

The material presented so far in this chapter has dealt exclusively with **single-phase AC power**, which implies a single sinusoidal source. However, most of the AC power used today is generated and distributed as **three-phase power**, which implies three sinusoidal sources, each out of phase with the other. The primary benefit is efficiency: The weight of the conductors and other components in a three-phase system is much lower than that in a single-phase system delivering the same amount of power. Further, while the power produced by a single-phase system has a pulsating nature (recall the results of section 7.1), a three-phase system can deliver a steady, constant supply of power. For example, later in this section it will be shown that a three-phase generator producing three **balanced voltages**—that is, voltages of equal amplitude and frequency displaced in phase by 120° —has the property of delivering constant instantaneous power.

The change to three-phase AC power systems from the early DC system proposed by Edison was due to a number of reasons: the efficiency resulting from transforming voltages up and down to minimize transmission losses over long distances, the ability to deliver constant power, a more efficient use of conductors, and the ability to provide starting torque for industrial motors.

Consider a three-phase source connected in a **wye (Y) configuration**, as shown in Figure 7.47. Each of the three voltages is 120° out of phase with the others,

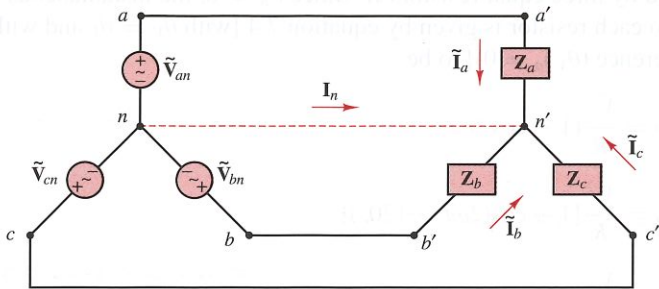


Figure 7.47 Balanced three-phase AC circuit

such that:



$$\begin{aligned}\tilde{V}_{an} &= \tilde{V}_{an} \angle 0^\circ \\ \tilde{V}_{bn} &= \tilde{V}_{bn} \angle -(120^\circ) \\ \tilde{V}_{cn} &= \tilde{V}_{cn} \angle (-240^\circ) = \tilde{V}_{cn} \angle 120^\circ\end{aligned}\quad \begin{array}{l} \text{Phase} \\ \text{voltages} \end{array}\quad (7.47)$$

If the three-phase source is *balanced*, then:

$$\tilde{V}_{an} + \tilde{V}_{bn} + \tilde{V}_{cn} = 0 \quad \text{Balanced phase voltages} \quad (7.48)$$

For three balanced **phase voltages** each separated by 120° , the phase amplitudes are also equal:

$$\tilde{V}_{an} = \tilde{V}_{bn} = \tilde{V}_{cn} = \tilde{V} \quad (7.49)$$

The result is the so-called **positive abc sequence**, as shown in Figure 7.48. In the wye configuration, the three phase voltages share a common *neutral node*, denoted by n .

It is also possible to define **line voltages** as the potential differences between lines aa' and bb' , lines aa' and cc' , and lines bb' and cc' . Each line voltage is related to the phase voltages by:

$$\begin{aligned}\tilde{V}_{ab} &= \tilde{V}_{an} - \tilde{V}_{bn} = \sqrt{3}\tilde{V} \angle 30^\circ \\ \tilde{V}_{bc} &= \tilde{V}_{bn} - \tilde{V}_{cn} = \sqrt{3}\tilde{V} \angle (-90^\circ) \\ \tilde{V}_{ca} &= \tilde{V}_{cn} - \tilde{V}_{an} = \sqrt{3}\tilde{V} \angle 150^\circ\end{aligned}\quad \begin{array}{l} \text{Line} \\ \text{voltages} \end{array}\quad (7.50)$$

It is instructive to note that the circuit of Figure 7.47 can be redrawn as shown in Figure 7.49, where it is clear that the three branches are in parallel.

When $Z_a = Z_b = Z_c = Z$, the wye load configuration is also balanced. When both the source and load networks are balanced, KCL requires that the current $\tilde{I}_n$ in the neutral line $n - n'$ be identically zero.

$$\tilde{I}_n = \tilde{I}_a + \tilde{I}_b + \tilde{I}_c = \frac{\tilde{V}_{an} + \tilde{V}_{bn} + \tilde{V}_{cn}}{Z} = 0 \quad (7.51)$$

Another important characteristic of a balanced three-phase power system is illustrated by a simplified version of Figure 7.49, where the balanced load impedances are replaced by three equal resistors R . Since $\theta_R = 0$, the instantaneous power $p(t)$ delivered to each resistor is given by equation 7.4 [with $\theta_V = \theta_I$ and with the freely chosen reference $(\theta_V)_a = 0$] to be:

$$\begin{aligned}p_a(t) &= \frac{\tilde{V}^2}{R} (1 + \cos 2\omega t) \\ p_b(t) &= \frac{\tilde{V}^2}{R} [1 + \cos(2\omega t - 120^\circ)] \\ p_c(t) &= \frac{\tilde{V}^2}{R} [1 + \cos(2\omega t + 120^\circ)]\end{aligned}\quad (7.52)$$

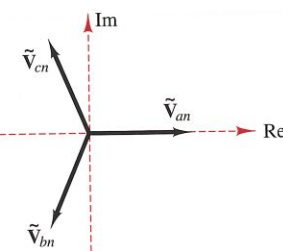


Figure 7.48 Positive, or *abc*, sequence for balanced three-phase voltages

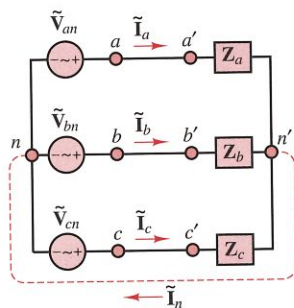


Figure 7.49 Balanced three-phase AC circuit (redrawn)

The total instantaneous power $p(t)$ delivered to the total load is the sum:

$$\begin{aligned}p(t) &= p_a(t) + p_b(t) + p_c(t) \\ &= \frac{\tilde{V}^2}{R} [3 + \cos 2\omega t + \cos(2\omega t - 120^\circ) + \cos(2\omega t + 120^\circ)] \\ &= \frac{3\tilde{V}^2}{R} = \text{constant!}\end{aligned}\quad (7.53)$$

It is worthwhile to verify that the sum of the three cosine terms is identically zero. (Hint: Consider the phasor sum of $e^{j(2\omega t)}$, $e^{j(2\omega t - \pi/3)}$, and $e^{j(2\omega t + \pi/3)}$.)

Thus, with the simplified balanced resistive load, the total power delivered to the load by the balanced three-phase source is constant. This is an extremely important result, for a very practical reason: Delivering power in a steady fashion (as opposed to the pulsating nature of single-phase power) reduces “wear and tear” on the source and load.

It is also possible to connect three AC sources in a **delta (Δ) configuration**, as shown in Figure 7.50 although it is rarely used in practice.

Balanced Wye Loads

These results for purely resistive loads can be generalized for any arbitrary balanced complex load. Consider again in Figure 7.47, where now the balanced load consists of three complex impedances:

$$Z_a = Z_b = Z_c = Z_y = |Z_y| \angle \theta \quad (7.54)$$

Because of the common neutral line $n - n'$, each impedance sees the corresponding phase voltage across itself. Therefore, since $\tilde{V}_{an} = \tilde{V}_{bn} = \tilde{V}_{cn}$, it is also true that $\tilde{I}_a = \tilde{I}_b = \tilde{I}_c$ and the phase angles of the currents will differ by $\pm 120^\circ$. Consequently, it is possible to compute the power for each phase from the phase voltage and the associated line current. Denote the complex power for each phase by S , where:

$$\begin{aligned}S &= P + jQ \\ &= \tilde{V}\tilde{I} \cos \theta + j\tilde{V}\tilde{I} \sin \theta\end{aligned}\quad (7.55)$$

The total real power delivered to the balanced wye load is $3P$, and the total reactive power is $3Q$. The total complex power S_T is:

$$\begin{aligned}S_T &= P_T + jQ_T = 3P + j3Q \\ &= \sqrt{(3P)^2 + (3Q)^2} \angle \theta\end{aligned}\quad (7.56)$$

The apparent power $|S_T|$ is:

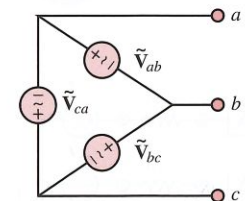
$$\begin{aligned}|S_T| &= 3\sqrt{(\tilde{V}\tilde{I})^2 \cos^2 \theta + (\tilde{V}\tilde{I})^2 \sin^2 \theta} \\ &= 3\tilde{V}\tilde{I}\end{aligned}\quad (7.57)$$

such that:

$$\begin{aligned}P_T &= |S_T| \cos \theta \\ Q_T &= |S_T| \sin \theta\end{aligned}\quad (7.58)$$

Balanced Delta Loads

It is also possible to assemble a balanced load in a delta configuration. A wye generator and a delta load are shown in Figure 7.51.



A delta-connected three-phase generator with line voltages V_{ab} , V_{bc} , V_{ca}

Figure 7.50 Delta configuration

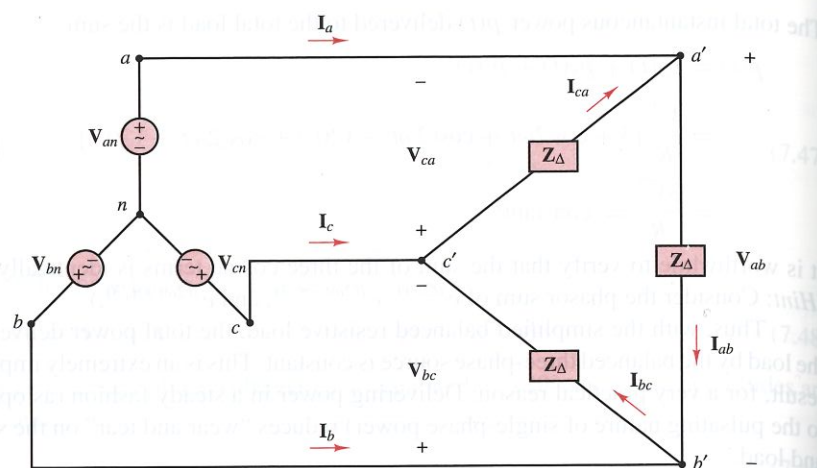


Figure 7.51 Balanced wye generators with balanced delta load

Note immediately that each impedance Z_{Δ} sees a corresponding line voltage, rather than a phase voltage. For example, the voltage across $Z_{c'a'}$ is $\tilde{V}_{ca}$. Thus, the three load currents are:

$$\begin{aligned}\tilde{I}_{ab} &= \frac{\tilde{V}_{ab}}{Z_{\Delta}} = \frac{\sqrt{3}\tilde{V}\angle(\pi/6)}{|Z_{\Delta}|\angle\theta} \\ \tilde{I}_{bc} &= \frac{\tilde{V}_{bc}}{Z_{\Delta}} = \frac{\sqrt{3}\tilde{V}\angle(-\pi/2)}{|Z_{\Delta}|\angle\theta} \\ \tilde{I}_{ca} &= \frac{\tilde{V}_{ca}}{Z_{\Delta}} = \frac{\sqrt{3}\tilde{V}\angle(5\pi/6)}{|Z_{\Delta}|\angle\theta}\end{aligned}\quad (7.59)$$

The relationship between a delta load and a wye load can be illustrated by determining the delta load Z_{Δ} that would draw the same amount of current as a wye load Z_y , assuming a given source voltage. Consider the circuits shown in Figures 7.47 and 7.51. For instance, the line current drawn in phase a by a wye load is:

$$(\tilde{I}_a)_y = \frac{\tilde{V}_{an}}{Z} = \frac{\tilde{V}}{|Z_y|}\angle(-\theta) \quad (7.60)$$

The current drawn by a delta load is:

$$\begin{aligned}(\tilde{I}_a)_{\Delta} &= \tilde{I}_{ab} - \tilde{I}_{ca} \\ &= \frac{\tilde{V}_{ab}}{Z_{\Delta}} - \frac{\tilde{V}_{ca}}{Z_{\Delta}} \\ &= \frac{1}{Z_{\Delta}}(\tilde{V}_{an} - \tilde{V}_{bn} - \tilde{V}_{cn} + \tilde{V}_{an}) \\ &= \frac{1}{Z_{\Delta}}(2\tilde{V}_{an} - \tilde{V}_{bn} - \tilde{V}_{cn}) \\ &= \frac{3\tilde{V}_{an}}{Z_{\Delta}} = \frac{3\tilde{V}}{|Z_{\Delta}|}\angle(-\theta)\end{aligned}\quad (7.61)$$

The two currents $(\tilde{I}_a)_{\Delta}$ and $(\tilde{I}_a)_y$ are equal if:

$$Z_{\Delta} = 3Z_y \quad (7.62)$$

This result also implies that a delta load will draw three times as much current and absorb three times as much power as a wye load with the same branch impedance.

EXAMPLE 7.16 Per-Phase Solution of Balanced Wye-Wye Circuit

Problem

Compute the power delivered to the load by the three-phase generator in the circuit shown in Figure 7.52.

Solution

Known Quantities: Source voltage, line resistance, load impedance.

Find: Power delivered to the load P_{load} .

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_{an} = 480\angle 0^\circ$ V rms; $\tilde{V}_{bn} = 480\angle(-2\pi/3)$ V rms; $\tilde{V}_{cn} = 480\angle(2\pi/3)$ V rms; $R_{\text{line}} = 2\ \Omega$; $R_{\text{neutral}} = 10\ \Omega$; $Z_y = R_o + jX_o = 2 + j4 = 4.47\angle 1.107$ Ω .

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians unless indicated otherwise.

Analysis: Since the circuit is balanced, $\tilde{V}_{n-n'} = 0$ and the current through the neutral line is zero. As a result, each phase has the structure shown in Figure 7.53. For example, the real power absorbed by the load in phase a is:

$$P_a = |\tilde{I}_a|^2 R_o$$

where

$$|\tilde{I}_a| = \left| \frac{\tilde{V}_a}{Z_y + R_{\text{line}}} \right| = \left| \frac{480\angle 0^\circ}{2 + j4 + 2} \right| = \left| \frac{480\angle 0^\circ}{5.66\angle(\pi/4)} \right| = 84.85 \text{ A rms}$$

and $P_a = (84.85 \text{ A})^2(2\ \Omega) = 14.4 \text{ kW}$. Since the circuit is balanced, the results for phases b and c are identical, such that:

$$P_{\text{load}} = 3P_a = 43.2 \text{ kW}$$

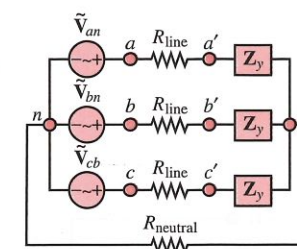


Figure 7.52

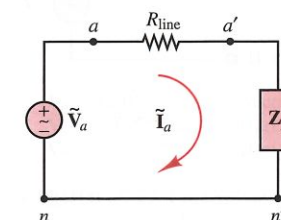


Figure 7.53 One phase of the three-phase circuit

EXAMPLE 7.17 Parallel Wye-Delta Load Circuit

Problem

Compute the power delivered to the wye-delta load by the three-phase generator in the circuit shown in Figure 7.54.

Solution

Known Quantities: Source voltage, line resistance, load impedance.



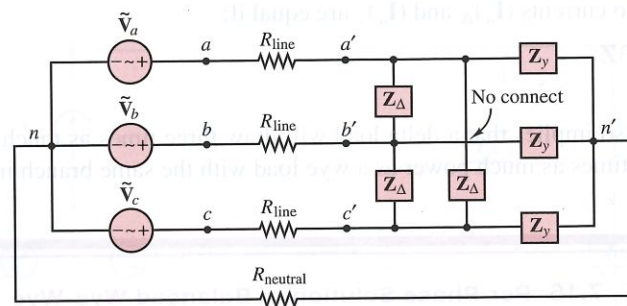


Figure 7.54 AC circuit with delta and wye loads

Find: Power delivered to the load P_{load} .

Schematics, Diagrams, Circuits, and Given Data: $\tilde{V}_{an} = 480\angle 0^\circ$ V rms; $\tilde{V}_{bn} = 480\angle(-2\pi/3)$ V rms; $\tilde{V}_{cn} = 480\angle(2\pi/3)$ V rms; $Z_y = 2 + j4 = 4.47\angle 1.107^\circ \Omega$; $Z_\Delta = 5 - j2 = 5.4\angle(-0.381)^\circ \Omega$; $R_{\text{line}} = 2 \Omega$; $R_{\text{neutral}} = 10 \Omega$.

Assumptions: All amplitudes are effective (rms) values. All angles are given in units of radians.

Analysis: First, convert the balanced delta load to an equivalent wye load, according to equation 7.62. Figure 7.55 illustrates the effect of this conversion.

$$Z_{\Delta-y} = \frac{Z_\Delta}{3} = \frac{5.4\angle(-0.381)^\circ}{3} = 1.667 - j0.667 = 1.8\angle(-0.381)^\circ \Omega$$

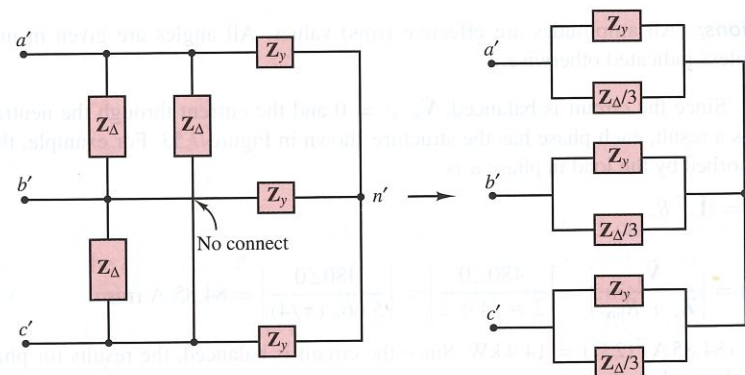


Figure 7.55 Conversion of delta load to equivalent wye load

Since the circuit is balanced, $\tilde{V}_{n-n'} = 0$ and the current through the neutral line is zero. The resulting per-phase circuit is shown in Figure 7.56. For example, the real power absorbed by the load in phase a is:

$$P_a = |\tilde{I}_a|^2 R_a = |\tilde{I}_a|^2 \text{Re}(Z_a)$$

where

$$Z_a = Z_y \parallel Z_{\Delta-y} = \frac{Z_y \times Z_{\Delta-y}}{Z_y + Z_{\Delta-y}} = 1.62 - j0.018 = 1.62\angle(-0.011)^\circ \Omega$$

The load current $|\tilde{I}_a|$ is:

$$|\tilde{I}_a| = \left| \frac{\tilde{V}_a}{Z_o + R_{\text{line}}} \right| = \left| \frac{480\angle 0^\circ}{1.62 + j0.018 + 2} \right| = 132.6 \text{ A rms}$$

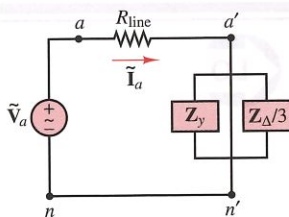


Figure 7.56 Per-phase circuit

Thus, $P_a = (132.6)^2 \times \text{Re}(Z_o) = 28.5 \text{ kW}$. Since the circuit is balanced, the results for phases b and c are identical, such that:

$$P_{\text{load}} = 3P_a = 85.5 \text{ kW}$$

CHECK YOUR UNDERSTANDING

Find the power lost in the line resistance shown in Example 7.16.

Compute the complex power S_o delivered to the balanced load of Example 7.16 if the lines have zero resistance and $Z_y = 1 + j3 \Omega$.

Show that the voltage across each branch of the wye load is equal to the corresponding phase voltage (e.g., the voltage across Z_a is $\tilde{V}_a$).

Prove that the sum of the instantaneous powers absorbed by the three branches in a balanced wye-connected load is constant and equal to $3\tilde{V}\tilde{I}\cos\theta$.

$$\text{Answers: } P_{\text{line}} = 43.2 \text{ kW}; S_o = 69.12 \text{ kW} + j207.4 \text{ kVA}$$

7.6 RESIDENTIAL WIRING; GROUNDING AND SAFETY

Common residential electric power service consists of a three-wire AC system supplied by the local power company. The three wires originate from a utility pole and consist of a neutral wire, which is connected to earth ground, and two "hot" wires. Each of the hot lines supplies 120 V rms to the residential circuits; the two lines are 180° out of phase, for reasons that will become apparent during the course of this discussion. The phasor line voltages, shown in Figure 7.57, are usually referred to by means of a subscript convention derived from the color of the insulation on the different wires: W for white (neutral), B for black (hot), and R for red (hot). This convention is adhered to uniformly.

The voltages across the hot lines are given by

$$\tilde{V}_B - \tilde{V}_R = \tilde{V}_{BR} = \tilde{V}_B - (-\tilde{V}_B) = 2\tilde{V}_B = 240\angle 0^\circ \quad (7.63)$$

Appliances such as electric stoves, air conditioners, and heaters are powered by the 240 V rms arrangement. On the other hand, lighting and all the electric outlets in the house used for small appliances are powered by a single 120 V rms line.

The use of 240 V rms service for appliances that require a substantial amount of power to operate is dictated by power transfer considerations. Consider the two circuits shown in Figure 7.58. In delivering the necessary power to a load, a lower line loss will be incurred with the 240 V rms wiring since the power loss in the lines (the I^2R loss, as it is commonly referred to) is directly related to the current required by the load. In an effort to minimize line losses, the size of the wires is increased for the lower-voltage case. This typically reduces the wire resistance by a factor of 2. In the top circuit, assuming $R_S/2 = 0.01 \Omega$, the current required by the 10-kW load is approximately 83.3 A while in the bottom circuit, with $R_S = 0.02 \Omega$, it is approximately one-half as much (41.7 A). (You should be able to verify that the approximate I^2R losses are 69.4 W in the top circuit and 34.7 W in the bottom circuit.)

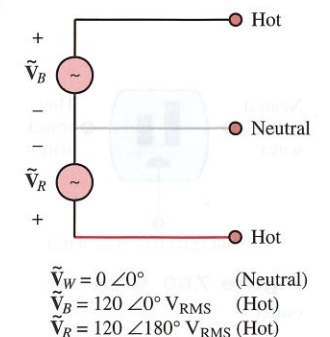


Figure 7.57 Line voltage convention for residential circuits

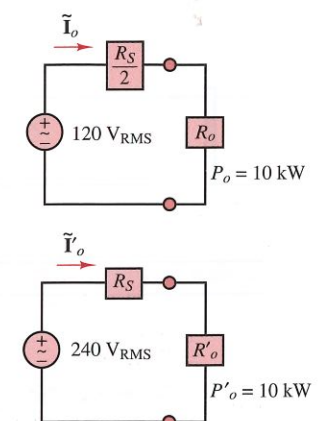


Figure 7.58 Line losses in 120- and 240-VAC circuits

Limiting the I^2R losses is important from the viewpoint of efficiency, besides reducing the amount of heat generated in the wiring for safety considerations. Figure 7.59 shows some typical wiring configurations for a home. Note that several circuits are wired and fused separately.

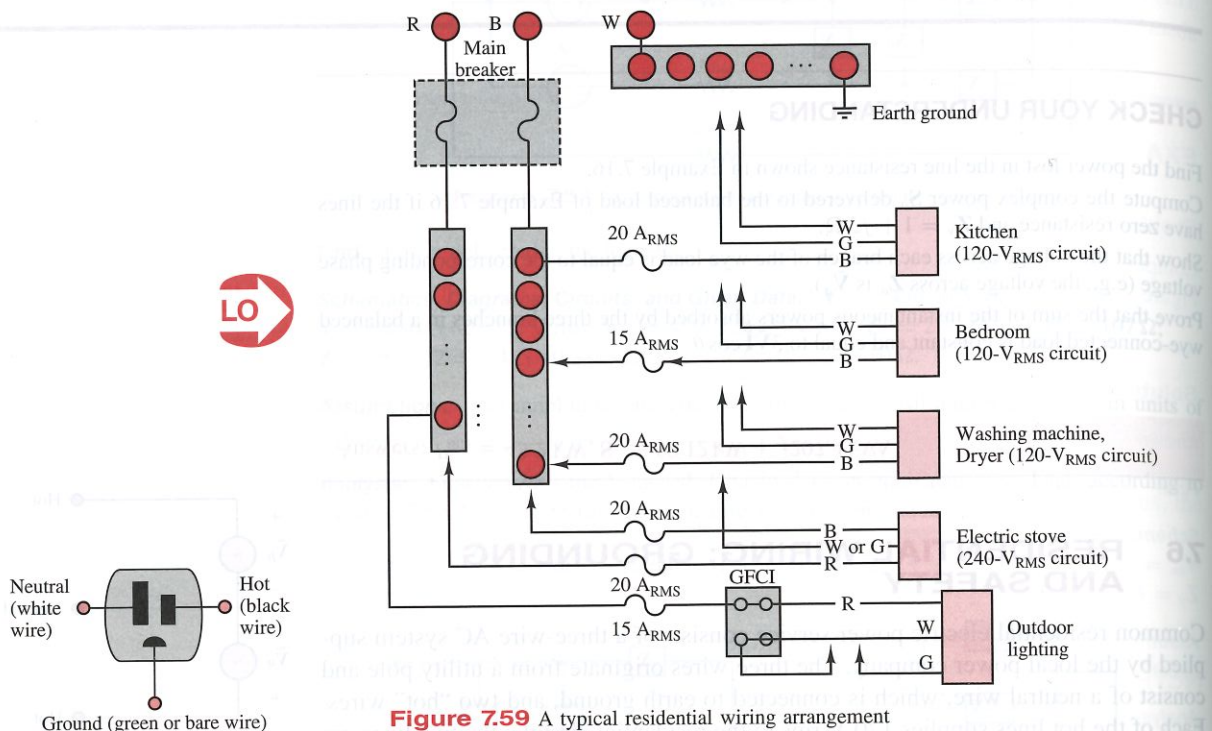


Figure 7.59 A typical residential wiring arrangement

Today, most homes have three wire connections to their outlets. The outlets appear as sketched in Figure 7.60. Then why are both the ground and neutral connections needed in an outlet? The answer to this question is *safety*: The ground connection is used to connect the chassis of the appliance to earth ground. Without this provision, the appliance chassis could be at any potential with respect to ground, possibly even at the hot wire's potential if a segment of the hot wire were to lose some insulation and come in contact with the inside of the chassis! Poorly grounded appliances can thus be a significant hazard. Figure 7.61 illustrates schematically how even though the chassis is intended to be insulated from the electric circuit, an unintended connection (represented by the dashed line) may occur, for example, because of corrosion or a loose mechanical connection. A path to ground might be provided by the body of a person touching the chassis with a hand. In the figure, such an undesired ground loop current is indicated by I_G . In this case, the ground current I_G would pass directly through the body to ground and could be harmful.

In some cases the danger posed by such undesired ground loops can be great, leading to death by electric shock. Figure 7.62 describes the effects of electric currents on an average male when the point of contact is dry skin. Particularly hazardous conditions are liable to occur whenever the natural resistance to current provided by the skin breaks down, as would happen in the presence of water. Thus, the danger presented to humans by unsafe electric circuits is very much dependent on the particular conditions—whenever water or moisture is present, the natural electrical

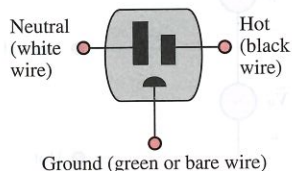


Figure 7.60 A three-wire outlet

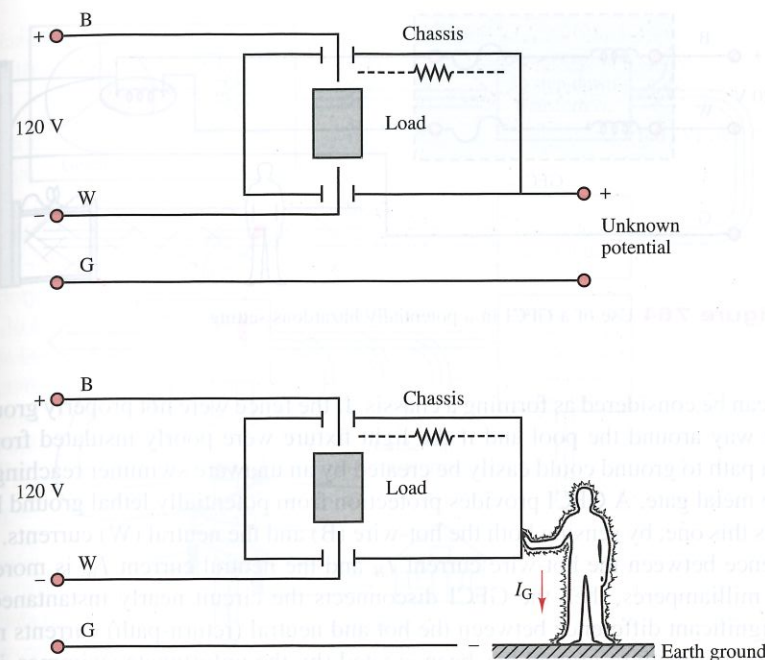


Figure 7.61 Unintended connection

resistance of dry skin, or of dry shoe soles, decreases dramatically, and even relatively low voltages can lead to fatal currents. Proper grounding procedures, such as those required by the National Electrical Code, help prevent fatalities due to electric shock. The **ground fault circuit interrupter**, labeled **GFCI** in Figure 7.59, is a special safety circuit used primarily with outdoor circuits and in bathrooms, where the risk of death by electric shock is greatest. Its application is best described by an example.

Consider the case of an outdoor pool surrounded by a metal fence, which uses an existing light pole for a post, as shown in Figure 7.63. The light pole and the metal

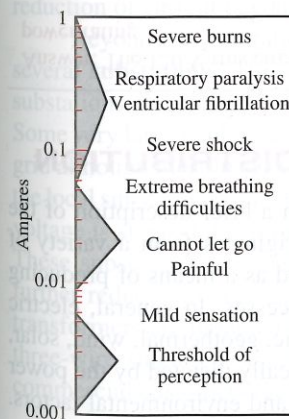


Figure 7.62 Physiological effects of electric currents

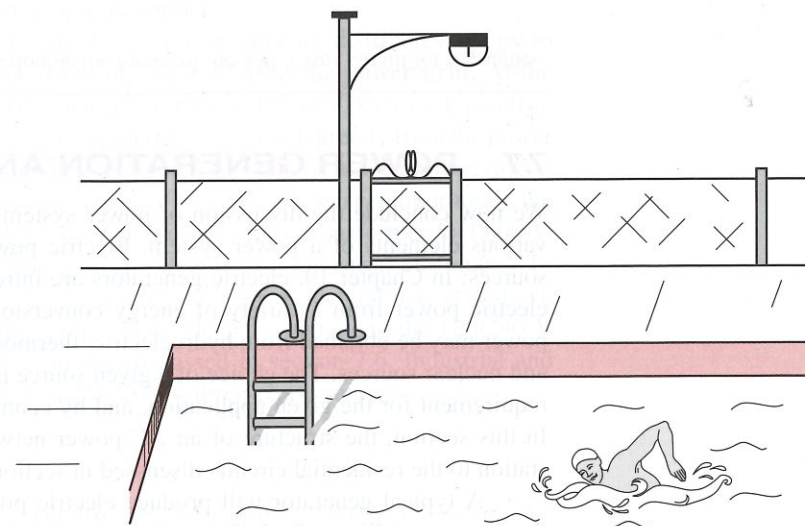


Figure 7.63 Outdoor pool

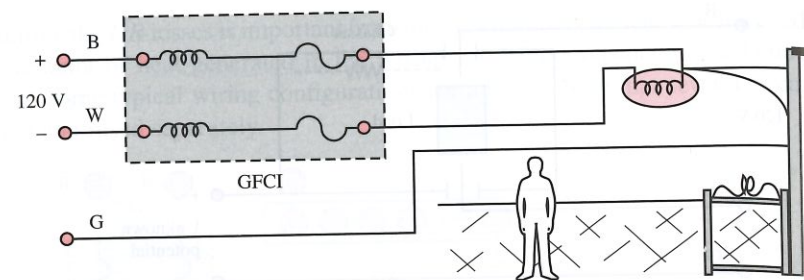


Figure 7.64 Use of a GFCI in a potentially hazardous setting

fence can be considered as forming a chassis. If the fence were not properly grounded all the way around the pool and if the light fixture were poorly insulated from the pole, a path to ground could easily be created by an unaware swimmer reaching, say, for the metal gate. A GFCI provides protection from potentially lethal ground loops, such as this one, by sensing both the hot-wire (B) and the neutral (W) currents. If the difference between the hot-wire current I_B and the neutral current I_W is more than a few milliamperes, then the GFCI disconnects the circuit nearly instantaneously. Any significant difference between the hot and neutral (return-path) currents means that a second path to ground has been created (by the unfortunate swimmer, in this example) and a potentially dangerous condition has arisen. Figure 7.64 illustrates the idea. GFCIs are typically resettable circuit breakers, so that one does not need to replace a fuse every time the GFCI circuit is enabled.

CHECK YOUR UNDERSTANDING

Use the circuits of Figure 7.58 to show that the I^2R losses will be higher for a 120-V service appliance than a 240-V service appliance if both have the same power usage rating.

Answer: The 120 V rms circuit has double the losses of the 240 V rms circuit for the same power rating.

7.7 POWER GENERATION AND DISTRIBUTION

We now conclude the discussion of power systems with a brief description of the various elements of a power system. Electric power originates from a variety of sources; in Chapter 19, electric generators are introduced as a means of producing electric power from a variety of energy conversion processes. In general, electric power may be obtained from hydroelectric, thermoelectric, geothermal, wind, solar, and nuclear sources. The choice of a given source is typically dictated by the power requirement for the given application, and by economic and environmental factors. In this section, the structure of an AC power network, from the power-generating station to the residential circuits discussed in section 7.6, is briefly outlined.

A typical generator will produce electric power at 18 kV rms, as shown in the diagram of Figure 7.65. To minimize losses along the conductors, the output of the generators is processed through a step-up transformer to achieve line voltages of

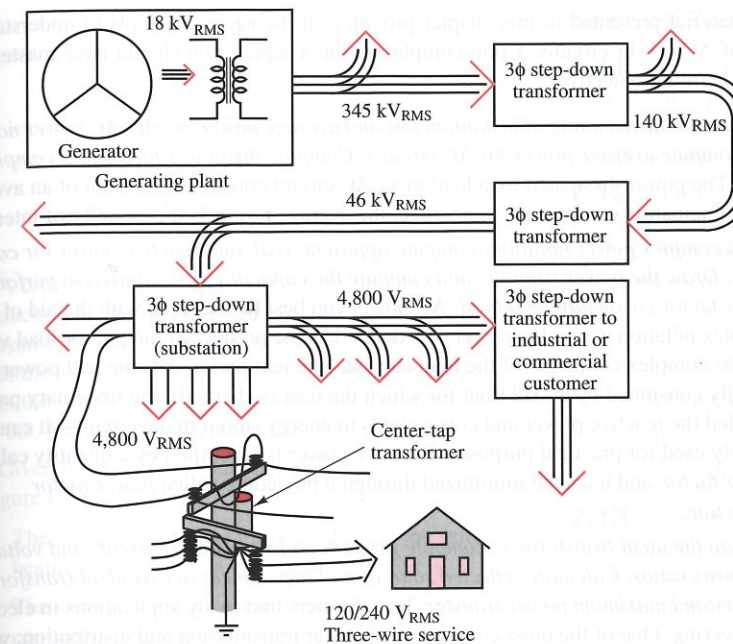


Figure 7.65 Structure of an AC power distribution network

hundreds of kilovolts (345 kV rms, in Figure 7.65). Without this transformation, the majority of the power generated would be lost in the **transmission lines** that carry the electric current from the power station.

The local electric company operates a power-generating plant that is capable of supplying several hundred megavolt-amperes (MVA) on a three-phase basis. For this reason, the power company uses a three-phase step-up transformer at the generation plant to increase the line voltage to around 345 kV rms. One can immediately see that at the rated power of the generator (in megavolt-amperes) there will be a significant reduction of current beyond the step-up transformer.

Beyond the generation plant, an electric power network distributes energy to several **substations**. This network is usually referred to as the **power grid**. At the substations, the voltage is stepped down to a lower level (10 to 150 kV rms, typically). Some very large loads (e.g., an industrial plant) may be served directly from the power grid although most loads are supplied by individual substations in the power grid. At the local substations (one of which you may have seen in your own neighborhood), the voltage is stepped down further by a three-phase step-down transformer to 4,800 V. These substations distribute the energy to residential and industrial customers. To further reduce the line voltage to levels that are safe for residential use, step-down transformers are mounted on utility poles. These drop the voltage to the 120/240-V three-wire single-phase residential service discussed in section 7.6. Industrial and commercial customers receive 460- and/or 208-V three-phase service.

Conclusion

Chapter 7 introduces the essential elements that permit the analysis of AC power systems. AC power is essential to all industrial activities and to the conveniences we are accustomed to in residential life. Virtually all engineers will be exposed to AC power systems in their careers,

and the material presented in this chapter provides all the necessary tools to understand the analysis of AC power circuits. Upon completing this chapter, you should have mastered the following learning objectives:

1. Understand the meaning of instantaneous and average power; master AC power notation, and compute average power for AC circuits. Compute the power factor of a complex load. The power dissipated by a load in an AC circuit consists of the sum of an average and a fluctuating component. In practice, the average power is the quantity of interest.
2. Learn complex power notation; compute apparent, real, and reactive power for complex loads. Draw the power triangle, and compute the capacitor size required to perform power factor correction on a load. AC power can best be analyzed with the aid of complex notation. Complex power S is defined as the product of the phasor load voltage and the complex conjugate of the load current. The real part of S is the real power actually consumed by a load (that for which the user is charged); the imaginary part of S is called the reactive power and corresponds to energy stored in the circuit—it cannot be directly used for practical purposes. Reactive power is quantified by a quantity called the power factor, and it can be minimized through a procedure called power factor correction.
3. Analyze the ideal transformer; compute primary and secondary currents and voltages and turns ratios. Calculate reflected sources and impedances across ideal transformers. Understand maximum power transfer. Transformers find many applications in electrical engineering. One of the most common is in power transmission and distribution, where the electric power generated at electric power plants is stepped “up” and “down” before and after transmission, to improve the overall efficiency of electric power distribution.
4. Learn three-phase AC power notation; compute load currents and voltages for balanced wye and delta loads. AC power is generated and distributed in three-phase form. Residential services are typically single-phase (making use of only one branch of the three-phase lines) while industrial applications are often served directly by three-phase power.
5. Understand the basic principles of residential electrical wiring, of electrical safety, and of the generation and distribution of AC power.

HOMEWORK PROBLEMS

Section 7.1: Instantaneous and Average Power

- 7.1** The heating element in a soldering iron has a resistance of $20\ \Omega$. Find the average power dissipated in the soldering iron if it is connected to a voltage source of 90 V rms .
- 7.2** A coffeemaker has a rated power of $1,000\text{ W}$ at 240 V rms . Find the resistance of the heating element.
- 7.3** A current source $i(t)$ is connected to a $50\text{-}\Omega$ resistor. Find the average power delivered to the resistor, given that $i(t)$ is
- a. $7\cos 100t\text{ A}$
 - b. $7\cos(100t - 30^\circ)\text{ A}$
 - c. $7\cos 100t - 3\cos(100t - 60^\circ)\text{ A}$
 - d. $7\cos 100t - 3\text{ A}$

- 7.4** Find the rms value of each of the following periodic currents:
- a. $\cos 200t + 3\cos 200t$
 - b. $\cos 10t + 2\sin 10t$
 - c. $\cos 50t + 1$
 - d. $\cos 30t + \cos(30t + \pi/6)$
- 7.5** A current of 2.5 A through a neon light advertisement is supplied by a 115 V rms voltage source. The current lags the voltage by 30° . Find the impedance of the light, the real power dissipated by it, and its power factor.
- 7.6** Compute the average power dissipated by the load seen by the voltage source in Figure P7.6. Let $\omega = 377\text{ rad/s}$, $\tilde{V}_s = 50\angle 0^\circ$, $R = 10\ \Omega$, $L = 0.08\text{ H}$, and $C = 200\ \mu\text{F}$.

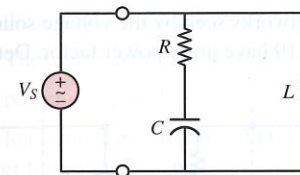


Figure P7.6

- 7.7** A drilling machine is driven by a single-phase induction machine connected to a 110 V rms supply. Assume that the machining operation requires 1 kW , that the tool machine has 90 percent efficiency, and that the supply current is 14 A rms with a power factor of 0.8. Find the AC machine efficiency.
- 7.8** Given the waveform of a voltage source shown in Figure P7.8, find:
- a. The steady DC voltage that would cause the same heating effect across a resistance.
 - b. The average current supplied to a $10\text{-}\Omega$ resistor connected across the voltage source.
 - c. The average power supplied to a $1\text{-}\Omega$ resistor connected across the voltage source.

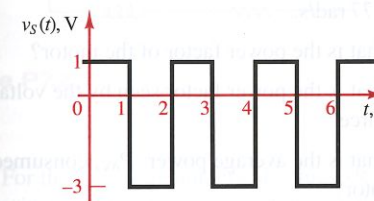


Figure P7.8

- 7.9** A current source $i(t)$ is connected to a $100\text{-}\Omega$ resistor. Find the average power delivered to the resistor, given that $i(t)$ is:
- a. $4\cos(100t)\text{ A}$
 - b. $4\cos(100t - 50^\circ)\text{ A}$
 - c. $4\cos(100t - 3)\cos(100t - 50^\circ)\text{ A}$
 - d. $4\cos(100t - 3)\text{ A}$
- 7.10** Find the rms value of each of the following periodic currents:
- a. $\cos(377t) + \cos(377t)\text{ A}$
 - b. $\cos(2t) + \sin(2t)\text{ A}$
 - c. $\cos(377t) + 1\text{ A}$
 - d. $\cos(2t) + \cos(2t + 135^\circ)\text{ A}$
 - e. $\cos(2t) + \cos(3t)\text{ A}$

Section 7.2: Complex Power

- 7.11** A current of 10 A rms results when a single-phase circuit is placed across a 220 V rms source. The current lags the voltage by 60° . Find the power dissipated by the circuit and the power factor.
- 7.12** A network is supplied by a 120 V rms , 60-Hz voltage source. An ammeter and a wattmeter indicate that 12 A rms is drawn from the source and 800 W are consumed by the network. Determine:
- a. The network power factor.
 - b. The network phase angle.
 - c. The network impedance.
 - d. The equivalent resistance and reactance of the network.
- 7.13** For the following numeric values, determine the average power, P , the reactive power, Q , and the complex power, S , of the circuit shown in Figure P7.13. Note: Phasor quantities are rms.
- a. $v_s(t) = 650\cos(377t)\text{ V}$
 $i_o(t) = 20\cos(377t - 10^\circ)\text{ A}$
 - b. $\tilde{V}_s = 460\angle 0^\circ\text{ V rms}$
 $\tilde{I}_o = 14.14\angle -45^\circ\text{ A rms}$
 - c. $\tilde{V}_s = 100\angle 0^\circ\text{ V rms}$
 $\tilde{I}_o = 8.6\angle -86^\circ\text{ A rms}$
 - d. $\tilde{V}_s = 208\angle -30^\circ\text{ V rms}$
 $\tilde{I}_o = 2.3\angle -63^\circ\text{ A rms}$

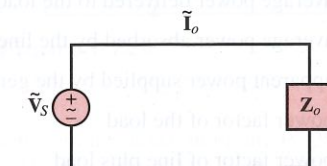


Figure P7.13

- 7.14** For the circuit of Figure P7.13, determine the power factor for the load Z_o and determine whether it is leading or lagging for the following conditions:
- a. $v_s(t) = 679\cos(\omega t + 15^\circ)\text{ V}$
 $i_o(t) = 20\cos(\omega t + 47^\circ)\text{ A}$
 - b. $v_s(t) = 163\cos(\omega t + 15^\circ)\text{ V}$
 $i_o(t) = 20\cos(\omega t - 22^\circ)\text{ A}$
 - c. $v_s(t) = 294\cos(\omega t)\text{ V}$
 $i_o(t) = 1.7\cos(\omega t + 175^\circ)\text{ A}$
 - d. $Z_o = (48 + j16)\ \Omega$
- 7.15** For the circuit of Figure P7.13, determine whether the load is capacitive or inductive, assuming:

a. $pf = 0.87$ (leading)

b. $pf = 0.42$ (leading)

c. $v_S(t) = 42 \cos(\omega t)$ V

$i_L(t) = 4.2 \sin(\omega t)$ A

d. $v_S(t) = 10.4 \cos(\omega t - 12^\circ)$ V

$i_L(t) = 0.4 \cos(\omega t - 12^\circ)$ A

7.16 For the circuit shown in Figure P7.16, assume $C = 265 \mu\text{F}$, $L = 25.55 \text{ mH}$, and $R = 10 \Omega$. Find the instantaneous real and reactive power if:

a. $v_S(t) = 120 \cos(377t)$ V (i.e., the frequency is 60 Hz)

b. $v_S(t) = 650 \cos(314t)$ V (i.e., the frequency is 50 Hz)

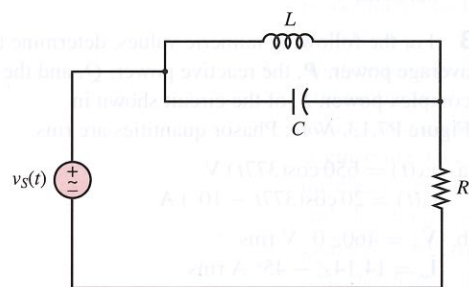


Figure P7.16

7.17 A load impedance, $Z_o = 10 + j3 \Omega$, is connected to a source with line resistance equal to 1Ω , as shown in Figure P7.17. Calculate the following values:

- The average power delivered to the load.
- The average power absorbed by the line.
- The apparent power supplied by the generator.
- The power factor of the load.
- The power factor of line plus load.

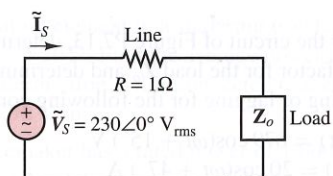


Figure P7.17

Section 7.3: Power Factor Correction

7.18 A single-phase motor draws 220 W at a power factor of 0.8 lagging when connected across a 240 V rms, 60-Hz source. A capacitor is connected in parallel with the load to produce a unity power factor. Determine the required capacitance.

7.19 The networks seen by the voltage sources in Figure P7.19 have unity power factor. Determine C_P and C_S .

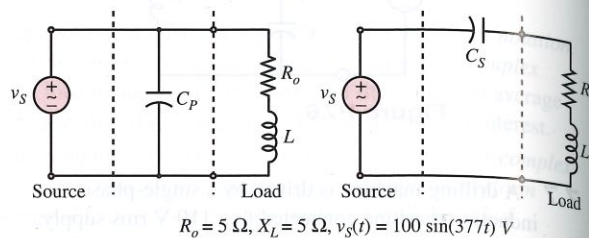


Figure P7.19

7.20 A 1,000-W electric motor is connected to a 120 V_{rms}, 60-Hz source. The power factor seen by the source is 0.8, lagging. To correct the pf to 0.95 lagging, a capacitor is placed in parallel with the motor. Calculate the current drawn from the source with and without the capacitor connected. Determine the value of the capacitor required to make the correction.

7.21 The motor inside a blender can be modeled as a resistance in series with an inductance, as shown in Figure P7.21. The wall socket source is modeled as an ideal 120 V rms voltage source in series with a 2-Ω output resistance. Assume the source frequency is $\omega = 377 \text{ rad/s}$.

- What is the power factor of the motor?
- What is the power factor seen by the voltage source?
- What is the average power, P_{AV} , consumed by the motor?
- What value of capacitor when placed in parallel with the motor will change the power factor seen by the voltage source to 0.9 lagging?

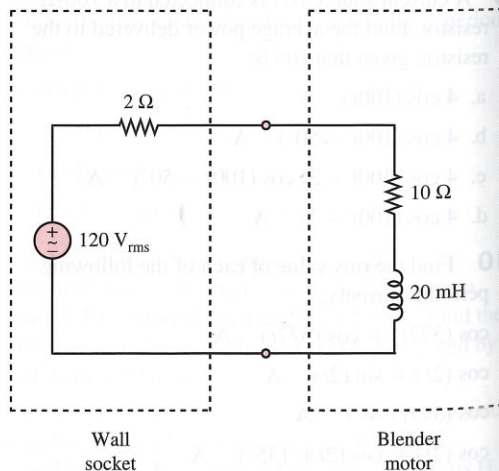


Figure P7.21

7.22 For the circuit shown in Figure P7.22, find:

- The Thévenin equivalent network seen by the load.
- The power dissipated by the load resistor.
- The load impedance that would result in maximum power transfer to the load.

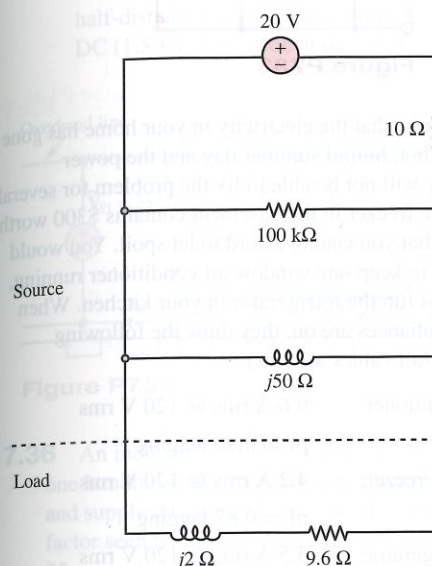


Figure P7.22

7.23 For the following numerical values, determine the capacitance to be placed in parallel with the load Z_o shown in Figure P7.13 that will result in a unity power factor seen by the voltage source. Assume $\omega = 377 \text{ rad/s}$.

- $\tilde{V}_s = 300\angle 0^\circ \text{ V rms}$, $\tilde{I}_o = 80\angle (-0.15\pi) \text{ A rms}$
- $\tilde{V}_s = 100\angle 0^\circ \text{ V rms}$, $\tilde{I}_o = 30\angle (-\pi/4) \text{ A rms}$
- $\tilde{V}_s = 12\angle (-\pi/4) \text{ V rms}$, $\tilde{I}_o = 3\angle (-\pi/2) \text{ A rms}$

7.24 For the circuit of Figure P7.13, determine the power factor of the load for each case listed below. Is it leading or lagging?

- $v_s(t) = 50 \cos(\omega t)$ V
 $i_o(t) = 20 \sin(\omega t + 1.2)$ A
- $v_s(t) = 110 \cos(\omega t + 0.1)$ V
 $i_o(t) = 10 \cos(\omega t - 0.1)$ A
- $Z_o = (20 + j5) \Omega$
- $Z_o = (20 - j5) \Omega$

7.25 For the circuit of Figure P7.13, determine whether the load Z_o is capacitive or inductive, if:

- its power factor is $pf = 0.76$ lagging.
- its power factor is $pf = 0.5$ (leading).
- $v_s(t) = 10 \cos(\omega t)$ V, $i_o(t) = \cos(\omega t)$ A.
- $v_s(t) = 100 \cos(\omega t)$ V, $i_o(t) = 12 \cos(\omega t + \pi/4)$ A.

7.26 Find the real and reactive power supplied by the voltage source shown in Figure P7.26 for $\omega = 5 \text{ rad/s}$ and $\omega = 15 \text{ rad/s}$. Let $v_s = 15 \cos(\omega t)$ V, $R = 5 \Omega$, $C = 0.1 \text{ F}$, $L_1 = 1 \text{ H}$, $L_2 = 2 \text{ H}$.

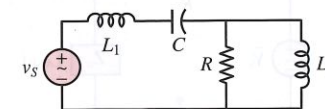


Figure P7.26

7.27 In Figure P7.27, assume $\tilde{V}_{S1} = 10\angle -\pi/4 \text{ V rms}$, $\tilde{V}_{S2} = 12\angle 0.8 \text{ V rms}$, $R_1 = 2 \Omega$, $R_2 = 3 \Omega$, $X_L = 4 \Omega$, and $X_C = -4 \Omega$. Find:

- The amplitude of the current supplied by each source.
- The total real power supplied by each source.

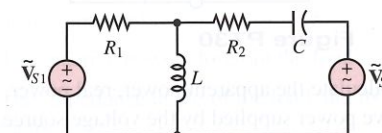


Figure P7.27

7.28 For the circuit shown in Figure P7.28, assume $f = 60 \text{ Hz}$, $\tilde{V}_s = 90\angle 0^\circ \text{ V rms}$, $R = 25 \Omega$, $X_L = 70 \Omega$, and $X_C = -8 \Omega$. Calculate:

- The capacitance C and the inductance L .
- The power factor seen by the voltage source.
- The new capacitance required to correct that power factor to unity.

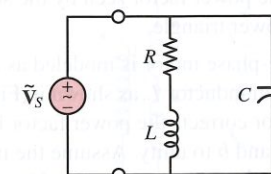


Figure P7.28

- 7.29** The load Z_o shown in Figure P7.29 consists of a $20\text{-}\Omega$ resistor in series with a 0.01-H inductor. Assuming $f = 60\text{ Hz}$, $R = 0.5\text{ }\Omega$, $\tilde{V}_s = 100\angle 0\text{ V rms}$. Calculate:
- a. The apparent power supplied by the voltage source.
 - b. The apparent power delivered to the load.
 - c. The power factor of the load.

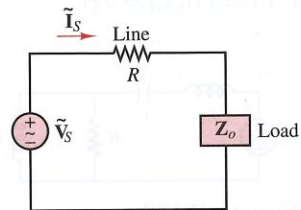


Figure P7.29

- 7.30** Calculate the real and reactive power of the load between terminals a and b in Figure P7.30. Assume $f = 60\text{ Hz}$, $\tilde{V}_s = 70\angle 0\text{ V rms}$, $R_s = 2\text{ }\Omega$, $R_o = 18\text{ }\Omega$, and $X_L = 5\text{ }\Omega$.

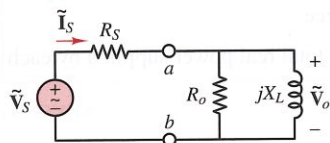


Figure P7.30

- 7.31** Calculate the apparent power, real power, and reactive power supplied by the voltage source shown in Figure P7.31. Draw the power triangle. Assume $f = 60\text{ Hz}$, $\tilde{V}_s = 70\angle 0\text{ V rms}$, $R = 18\text{ }\Omega$, $C = 50\text{ }\mu\text{F}$, and $L = 0.001\text{ H}$.

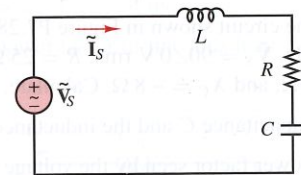


Figure P7.31

- 7.32** Refer to problem 7.31 and determine the capacitance needed in parallel with the voltage source to correct the power factor seen by the source to 0.95. Draw the power triangle.
- 7.33** A single-phase motor is modeled as a resistor R in series with an inductor L as shown in Figure P7.33. The capacitor corrects the power factor between terminals a and b to unity. Assume the meters shown are ideal and $f = 50\text{ Hz}$, $V = 220\text{ V rms}$, $I = 20\text{ A rms}$, and $I_1 = 25\text{ A rms}$. Find the capacitor value.

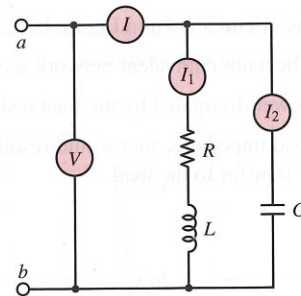


Figure P7.33

- 7.34** Suppose that the electricity in your home has gone out on a hot, humid summer day and the power company will not be able to fix the problem for several days. The freezer in the basement contains \$300 worth of food that you cannot afford to let spoil. You would also like to keep one window air conditioner running, as well as run the refrigerator in your kitchen. When these appliances are on, they draw the following currents (all values are rms):

Air conditioner:	9.6 A rms @ 120 V rms
	pf = 0.90 lagging
Freezer:	4.2 A rms @ 120 V rms
	pf = 0.87 lagging
Refrigerator:	3.5 A rms @ 120 V rms
	pf = 0.80 lagging

In the worst-case scenario, how much power must an emergency generator supply?

- 7.35** The French TGV high-speed train absorbs 11 MW at 300 km/h (186 mi/h). The power supply module shown in Figure P7.35 consists of two 25-kV rms single-phase AC power stations connected at the same overhead line, one at each end of the module. For the return circuits, the rail is used. The train is also designed to operate at a low speed with 1.5-kV DC in railway stations or under the old electrification lines. The natural (average) power factor in the AC operation is 0.8. Assume that the equivalent specific resistance of the overhead line is $0.2\text{ }\Omega/\text{km}$ and that the rail resistance can be neglected. Find:

- a. A simple circuit model for the system.
- b. The locomotive's current in the condition of a 10 percent voltage drop.
- c. The reactive power supplied by the power stations.
- d. The supplied real power, overhead line losses, and maximum distance between two power stations supplied in the condition of a 10 percent voltage drop when the train is located at the half-distance between the stations.

- e. Overhead line losses in the condition of a 10 percent voltage drop when the train is located at the half-distance between the stations, assuming pf = 1. (The French TGV is designed with a state-of-the-art power compensation system.)
- f. The maximum distance between the two power stations supplied in the condition of a 10 percent voltage drop when the train is located at the half-distance between the stations, assuming the DC (1.5-kV) operation at one-quarter power.

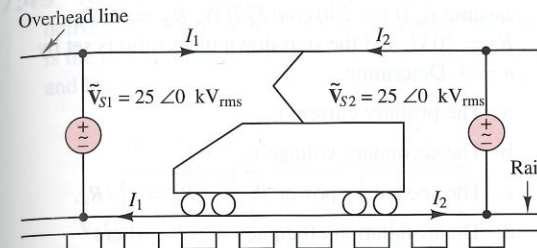


Figure P7.35

- 7.36** An industrial assembly hall is continuously lit by one hundred 40-W mercury vapor lamps in parallel and supplied by a 120 V rms, 60-Hz source. The power factor seen by the source is 0.65, which is so low that a 25 percent penalty is applied at billing. If the average price of 1 kWh is \$0.05 and the average cost of a capacitor is \$50 per mF, compute how long it will take before the billing penalty equals the cost of the capacitor needed to correct the power factor to 0.85.

- 7.37** Refer to Problem 7.36 and assume that each lamp is now available with a compensating capacitor in parallel with the original lamp. Find:
- a. The compensating capacitor value for unity power factor seen by the source.
 - b. The maximum number of additional lamps that can be installed without exceeding the original current supplied by the source when using uncompensated lamps.

- 7.38** The voltage and current supplied by a source to a load are:

$$\tilde{V}_s = 7\angle 0.873\text{ V rms} \quad \tilde{I}_s = 13\angle (-0.349)\text{ A rms}$$

Determine:

- a. The real power consumed as work and dissipated as heat in the load.
- b. The reactive power stored in the load.
- c. The impedance angle of the load and its power factor.

- 7.39** Determine the real power dissipated and the reactive power stored in each of the impedances shown in Figure P7.39. Assume:

$$\tilde{V}_{s1} = \frac{170}{\sqrt{2}}\angle 0\text{ V rms}$$

$$\tilde{V}_{s2} = \frac{170}{\sqrt{2}}\angle \pi/2\text{ V rms}$$

$$\omega = 377\text{ rad/s}$$

$$Z_1 = 0.7\angle \frac{\pi}{6}\text{ }\Omega$$

$$Z_2 = 1.5\angle 0.105\text{ }\Omega$$

$$Z_3 = 0.3 + j0.4\text{ }\Omega$$

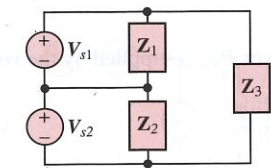


Figure P7.39

- 7.40** The following are supplied by a source to a load:
- $$\tilde{V}_s = 170\angle (-0.157)\text{ V rms} \quad \tilde{I}_s = 13\angle 0.28\text{ A rms}$$

Determine:

- a. The real power consumed as work and dissipated as heat in the load.
- b. The reactive power stored in the load.
- c. The impedance angle of the load and its power factor.

Section 7.4: Transformers

- 7.41** A center-tapped transformer has the schematic representation shown in Figure P7.41. The primary-side voltage is stepped down to two secondary-side voltages. Assume that each secondary supplies a 7-kW resistive load and that the primary is connected to 100 V rms. Find:

- a. The primary power.
- b. The primary current.

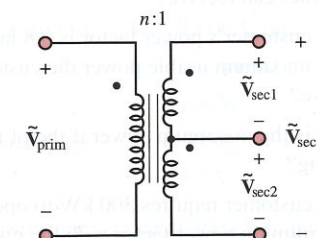


Figure P7.41

7.42 A center-tapped transformer has the schematic representation shown in Figure P7.41. The primary-side voltage is stepped down to a secondary-side voltage $\tilde{V}_{\text{sec}}$ by a ratio of $n:1$. On the secondary side, $\tilde{V}_{\text{sec1}} = \tilde{V}_{\text{sec2}} = \frac{1}{2} \tilde{V}_{\text{sec}}$.

- If $\tilde{V}_{\text{prim}} = 220 \angle 0^\circ$ V rms and $n = 11$, find $\tilde{V}_{\text{sec}}$, $\tilde{V}_{\text{sec1}}$, and $\tilde{V}_{\text{sec2}}$.
- What must n be if $\tilde{V}_{\text{prim}} = 110 \angle 0^\circ$ V rms and we desire $|\tilde{V}_{\text{sec2}}|$ to be 5 V rms?

7.43 For the circuit shown in Figure P7.43, assume that $\tilde{V}_g = 80 \angle 0^\circ$ V rms, $R_g = 2 \Omega$, and $R_o = 12 \Omega$. Assume an ideal transformer. Find:

- The equivalent resistance seen by the voltage source.
- The power P_{source} supplied by the voltage source.

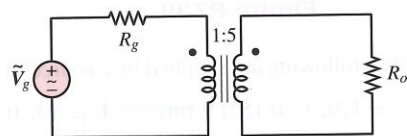


Figure P7.43

7.44 Refer to Problem 7.43 and find:

- The power P_{load} consumed by R_o .
- The installation efficiency $P_{\text{load}}/P_{\text{source}}$.
- The load R_o that results in maximum power transfer to the load.

7.45 An ideal transformer is rated to deliver 460 kVA at 380 V rms to a customer, as shown in Figure P7.45.

- How much current can the transformer supply to the customer?
- If the customer's load is purely resistive (i.e., if $\text{pf} = 1$), what is the maximum power that the customer can receive?
- If the customer's power factor is 0.8 lagging, what is the maximum usable power the customer can receive?
- What is the maximum power if the pf is 0.7 lagging?
- If the customer requires 300 kW to operate, what is the minimum power factor with the given size transformer?

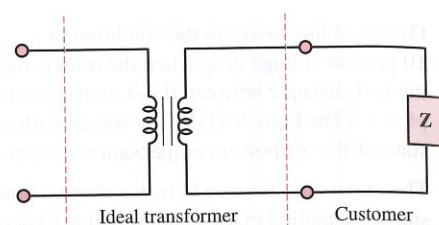


Figure P7.45

7.46 For the ideal transformer shown in Figure P7.46, assume $v_{\text{in}}(t) = 240 \cos(377t)$ V, $R_{\text{in}} = 50 \Omega$, $R_o = 20 \Omega$, and the step-down turns ratio is set by $n = 3$. Determine:

- The primary current i_{in} .
- The secondary voltage v_o .
- The secondary power $P_o = i_o^2 R_o = v_o^2 / R_o$.
- The installation efficiency P_{in}/P_o , where $P_{\text{in}} = i_{\text{in}}^2 R_{\text{in}}$.

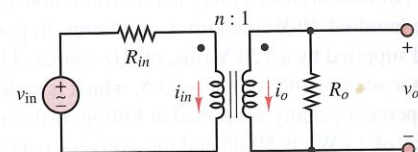


Figure P7.46

7.47 For Figure P7.47, assume the transformer is ideal. Find the step-down turns ratio $M = n$ that provides maximum power transfer to R_o . Let $R_{\text{in}} = 1,200 \Omega$, $R_o = 100 \Omega$, and $v_{\text{in}}(t) = V_{\text{pk}} \cos(\omega t)$.

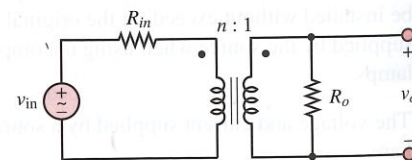


Figure P7.47

7.48 Consider the 8- Ω resistor shown in Figure P7.48 to be the load. Assume $\tilde{V}_g = 110 \angle 0^\circ$ V rms and a variable turns ratio n . What value of n results in maximum power (a) dissipated by the load? (b) supplied by the voltage source? What value of n results in maximum power transfer efficiency from source to load?

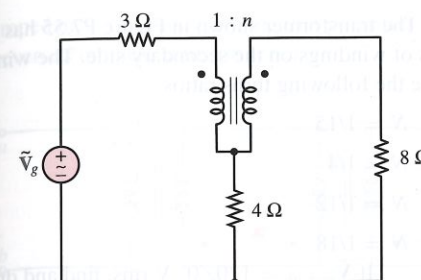


Figure P7.48

7.49 Assume the transformer shown in Figure P7.49 delivers 70 A rms at 90 V rms to a resistive load. What is the power transfer efficiency between voltage source and load? Let $R_s = 2 \Omega$, $X_{C1} = -10 \Omega$, $X_{C2} = -5 \Omega$.

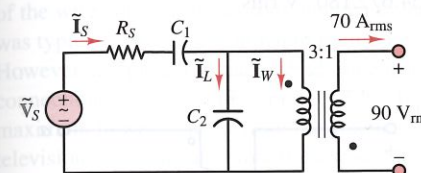


Figure P7.49

7.50 A method for determining the equivalent network of a non-ideal transformer consists of two tests: the open-circuit test and the short-circuit test. The open-circuit test, shown in Figure P7.50(a), is usually done by applying rated voltage to the primary side of the transformer while leaving the secondary side open.

The current into the primary side is measured, as is the power dissipated. The short-circuit test, shown in Figure P7.50(b), is performed by increasing the primary voltage until rated current is going into the transformer while the secondary side is short-circuited. The current into the transformer, the applied voltage, and the power dissipated are measured.

The equivalent circuit of a transformer is shown in Figure P7.50(c), where r_w and L_w represent the winding resistance and inductance, respectively, and r_c and L_c represent the losses in the core of the transformer and the inductance of the core. The ideal transformer is also included in the model.

With the open-circuit test, we may assume that $\tilde{I}_{\text{primary}} = \tilde{I}_{\text{secondary}} = 0$. Then all the current that is measured is directed through the parallel combination of r_c and L_c . We also assume that $|r_c| \ll |j\omega L_c|$ is much greater than $r_w + j\omega L_w$. Using these assumptions and the open-circuit test data, we can find the resistance r_c and the inductance L_c .

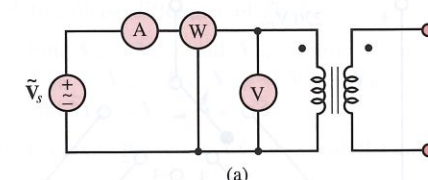
In the short-circuit test, we assume that $\tilde{V}_{\text{secondary}}$ is zero, so that the voltage on the primary side of the ideal transformer is also zero, causing no current through the $r_c \parallel L_c$ parallel combination. Using this

assumption with the short-circuit test data, we are able to find the resistance r_w and inductance L_w .

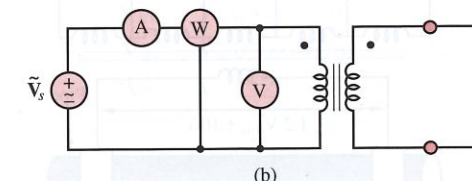
The following test data was measured by the meters indicated in Figure P7.50(a) and (b):

Open-circuit test: $\tilde{V} = 241$ V rms
 $\tilde{I} = 0.95$ A rms
 $P = 32$ W
 Short-circuit test: $\tilde{V} = 5$ V rms
 $\tilde{I} = 5.25$ A rms
 $P = 26$ W

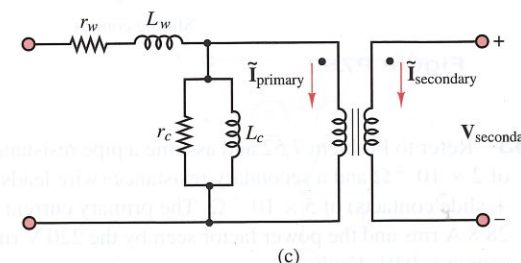
Both tests were made at $\omega = 377$ rad/s. Use the data to determine the equivalent network of the non-ideal transformer.



(a)



(b)



(c)

Figure P7.50

7.51 Use the methods outlined in Problem 7.50 and the following data to find the equivalent network of a non-ideal transformer.

Open-circuit test: $\tilde{V} = 4,600$ V rms
 $\tilde{I} = 0.7$ A rms
 $P = 200$ W
 Short-circuit test: $P = 50$ W
 $\tilde{V} = 5.2$ V rms

The transformer is rated at 460 kVA, and tests are performed at 60 Hz.

7.52 A method of thermal treatment for a steel pipe is to heat the pipe by the Joule effect, caused when a current is directed through the pipe. In most cases, a low-voltage, high-current transformer is used to deliver the current through the pipe. In this problem, we consider a single-phase transformer at 220 V rms, which delivers 1.2 V rms. Because of the pipe's resistance variation with temperature, a secondary voltage regulation is needed in the range of 10 percent, as shown in Figure P7.52. The voltage regulation is obtained with five different slots in the primary winding (high-voltage regulation). Assuming that the secondary coil has two turns, find the number of turns for each slot.

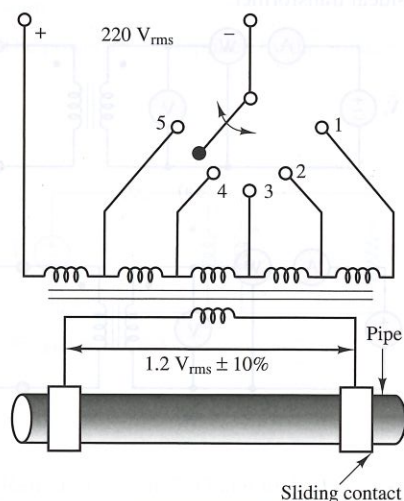


Figure P7.52

7.53 Refer to Problem 7.52 and assume a pipe resistance of $2 \times 10^{-4} \Omega$ and a secondary resistance (wire leads + slide contacts) of $5 \times 10^{-5} \Omega$. The primary current is 28.8 A rms and the power factor seen by the 220 V rms source is 0.91. Find:

- The slot number.
- The secondary reactance.
- The power transfer efficiency.

7.54 A single-phase transformer used for street lighting (high-pressure sodium discharge lamps) converts 6 kV rms to 230 V rms with an efficiency of 0.95. Assuming the power factor seen by the high voltage source is 0.8 and the primary apparent power is 30 kVA, find:

- The secondary current.
- The transformer turns ratio N .

7.55 The transformer shown in Figure P7.55 has several sets of windings on the secondary side. The windings have the following turns ratios:

- $N = 1/15$
- $N = 1/4$
- $N = 1/12$
- $N = 1/18$

If $\tilde{V}_{\text{primary}} = 120\angle 0^\circ$ V rms, find and draw the connections that will allow you to produce the following secondary voltages:

- $24.67\angle 0^\circ$ V rms
- $36.67\angle 0^\circ$ V rms
- $18\angle 0^\circ$ V rms
- $54.67\angle 180^\circ$ V rms

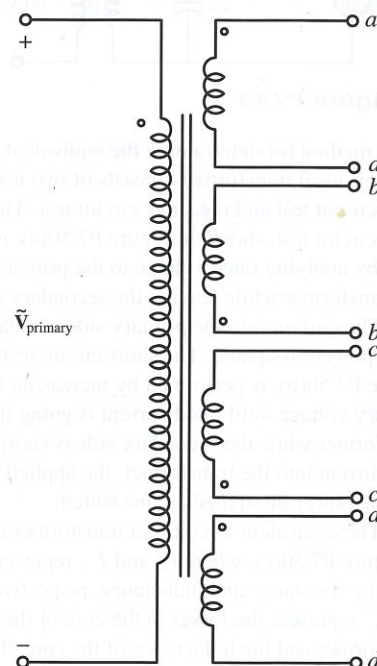


Figure P7.55

7.56 The circuit in Figure P7.56 shows the use of ideal transformers for impedance matching. You have a limited choice of turns ratios among available transformers. Suppose you can find transformers with turns ratios of 2:1, 7:2, 120:1, 3:2, and 6:1. If Z_o is $475\angle -25^\circ \Omega$ and Z_{ab} must be $267\angle -25^\circ$, find the combination of transformers that will provide this

impedance. (You may assume that polarities are easily reversed on these transformers.)

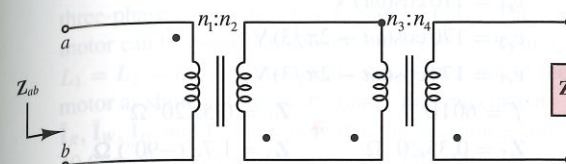


Figure P7.56

7.57 Before cable TV was generally available, TV networks broadcast their signals wirelessly. Large antennas were often installed atop residential homes to improve the reception of these signals. The impedance of the wire connecting the roof antenna to the TV set was typically 300Ω , as shown in Figure P7.57(a). However, a typical TV had a $75\text{-}\Omega$ impedance connection, as shown in Figure P7.57(b). To achieve maximum power transfer from the antenna to the television set, an ideal transformer was placed between the antenna and the TV, as shown in Figure P7.57(c). What is the turns ratio, $N = 1/n$, needed to obtain maximum power transfer?

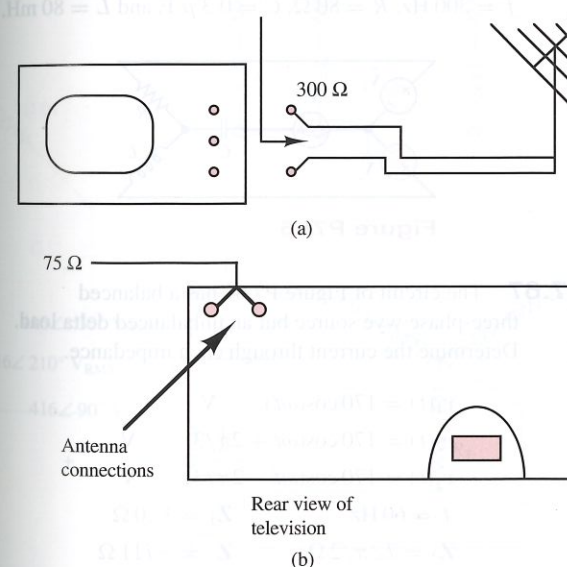


Figure P7.57

Section 7.5: Three-Phase Power

7.58 The magnitude of the phase voltage of a balanced three-phase wye system is 208 V rms. Express each phase and line voltage in both polar and rectangular coordinates.

7.59 The phase currents in a four-wire wye-connected load, such as that shown in Figure 7.49, are:

$$\tilde{I}_{an} = 22\angle 0^\circ \text{ A rms} \quad \tilde{I}_{bn} = 10\angle \frac{2\pi}{3} \text{ A rms} \quad \tilde{I}_{cn} = 15\angle \frac{\pi}{4}$$

Determine the current in the neutral wire.

7.60 Each voltage source shown in Figure P7.60 has a relative phase difference of $2\pi/3$.

- Find $\tilde{V}_{RW}$, $\tilde{V}_{WB}$, and $\tilde{V}_{BR}$, where $\tilde{V}_{RW} = \tilde{V}_R - \tilde{V}_W$, $\tilde{V}_{WB} = \tilde{V}_W - \tilde{V}_B$, and $\tilde{V}_{BR} = \tilde{V}_B - \tilde{V}_R$.
- Compare the results of part a with the calculations:

$$\begin{aligned} \tilde{V}_{RW} &= \tilde{V}_R \sqrt{3} \angle (-\pi/6) \\ \tilde{V}_{WB} &= \tilde{V}_W \sqrt{3} \angle (-\pi/6) \\ \tilde{V}_{BR} &= \tilde{V}_B \sqrt{3} \angle (-\pi/6) \end{aligned}$$

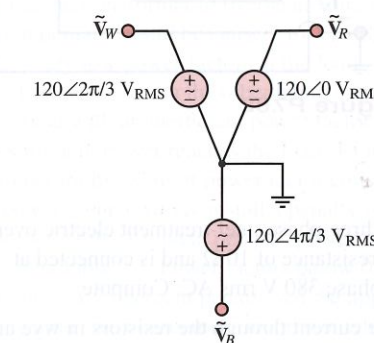


Figure P7.60

7.61 For the three-phase network shown in Figure P7.61, find the current in each wire and the real power consumed by the wye network. Let $\tilde{V}_R = 110\angle 0^\circ$ V rms, $\tilde{V}_W = 110\angle 2\pi/3$ V rms, and $\tilde{V}_B = 110\angle 4\pi/3$ V rms. $R = 50 \Omega$, $L = 120$ mH, $C = 133 \mu\text{F}$, $f = 60$ Hz.

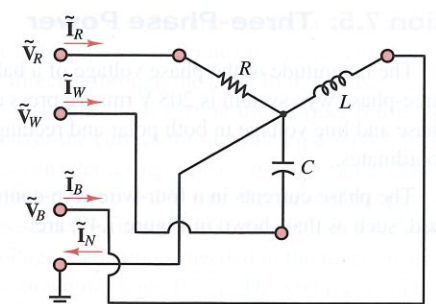


Figure P7.61

7.62 For the three-phase network shown in Figure P7.62, find the current in each wire and the real power consumed by the wye network. Let $\tilde{V}_R = 170\angle 0^\circ$ V rms, $\tilde{V}_W = 170\angle 2\pi/3$ V rms, and $\tilde{V}_B = 170\angle 4\pi/3$ V rms.

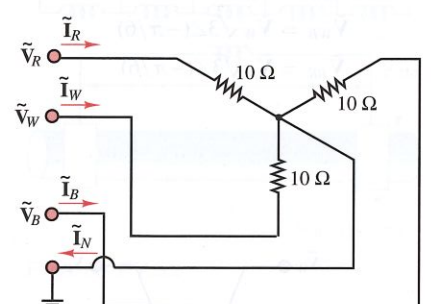


Figure P7.62

7.63 A three-phase steel-treatment electric oven has a phase resistance of $10\ \Omega$ and is connected at three-phase 380 V rms AC. Compute

- The current through the resistors in wye and delta connections.
- The power of the oven in wye and delta connections.

7.64 A naval in-board synchronous generator has an apparent power of 50 kVA and supplies a three-phase network of 380 V rms. Compute the phase currents, the real power, and the reactive power if:

- The power factor is 0.85.
- The power factor is 1.

7.65 The three-phase circuit shown in Figure P7.65 has a balanced wye source but an unbalanced wye load.

$$\begin{aligned} v_{s1} &= 170 \cos(\omega t) \text{ V} \\ v_{s2} &= 170 \cos(\omega t + 2\pi/3) \text{ V} \\ v_{s3} &= 170 \cos(\omega t - 2\pi/3) \text{ V} \\ f &= 60 \text{ Hz} \quad Z_1 = 0.5\angle 20^\circ \ \Omega \\ Z_2 &= 0.35\angle 0^\circ \ \Omega \quad Z_3 = 1.7\angle (-90^\circ) \ \Omega \end{aligned}$$

Determine the current through Z_1 , using the following methods:

- Mesh analysis.
- Superposition.

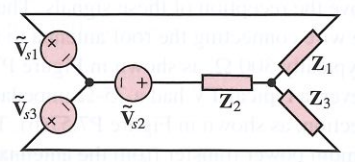


Figure P7.65

7.66 Determine the current through R shown in Figure P7.66. Assume: $\tilde{V}_1 = 150\angle 0^\circ$ V rms, $\tilde{V}_2 = 150\angle 2\pi/3$ V rms, $\tilde{V}_3 = 150\angle 4\pi/3$ V rms, $f = 300$ Hz, $R = 80\ \Omega$, $C = 0.3\ \mu\text{F}$, and $L = 80$ mH.

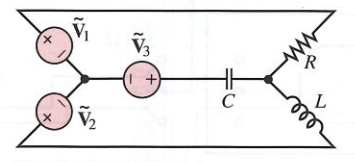


Figure P7.66

7.67 The circuit of Figure P7.67 has a balanced three-phase wye source but an unbalanced delta load. Determine the current through each impedance.

$$\begin{aligned} v_1(t) &= 170 \cos(\omega t) \text{ V} \\ v_2(t) &= 170 \cos(\omega t + 2\pi/3) \text{ V} \\ v_3(t) &= 170 \cos(\omega t - 2\pi/3) \text{ V} \\ f &= 60 \text{ Hz} \quad Z_1 = 3\angle 0^\circ \ \Omega \\ Z_2 &= 7\angle \pi/2 \ \Omega \quad Z_3 = -j11 \ \Omega \end{aligned}$$

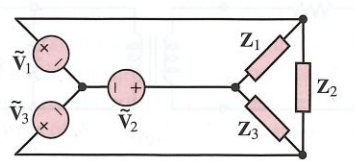


Figure P7.67

7.68 If we model each winding of a three-phase motor like the circuit shown in Figure P7.68(a) and connect the windings as shown in Figure P7.68(b), we have the three-phase circuit shown in Figure P7.68(c). The motor can be constructed so that $R_1 = R_2 = R_3$ and $L_1 = L_2 = L_3$, as is the usual case. If we connect the motor as shown in Figure P7.68(c), find the currents $\tilde{I}_R$, $\tilde{I}_W$, $\tilde{I}_B$, and $\tilde{I}_N$, assuming that the resistances are $40\ \Omega$ each and each inductance is 5 mH. The frequency of each source is 60 Hz.

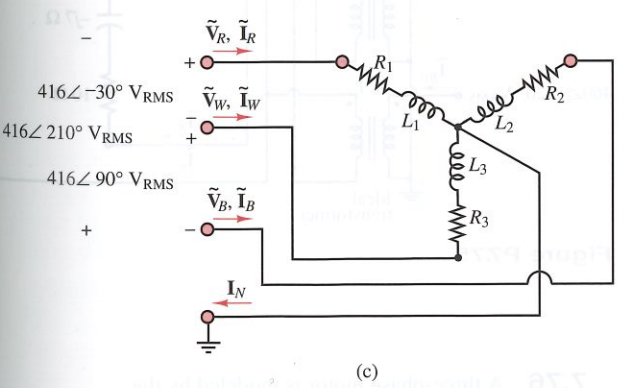
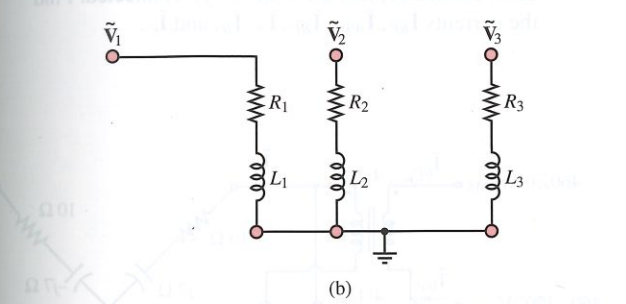
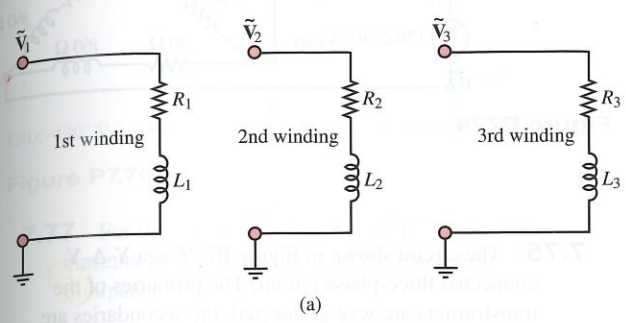


Figure P7.68

7.69 With reference to the motor of Problem 7.67,

- How much power (in watts) is delivered to the motor?

- What is the motor's power factor?
- Why is it common in industrial practice *not* to connect the ground lead to motors of this type?

7.70 In general, a three-phase induction motor is designed for wye connection operation. However, for short-time operation, a delta connection can be used at the nominal wye voltage. Find the ratio between the power delivered to the same motor in the wye and delta connections.

7.71 A residential four-wire system supplies power at 240 V rms to the following single-phase appliances: On the first phase, there are ten 60-W bulbs. On the second phase, there is a 1-kW vacuum cleaner with a power factor of 0.9. On the third phase, there are ten 23-W compact fluorescent lamps with a power factor of 0.61. Find:

- The current in the neutral wire.
- The real, reactive, and apparent power for each phase.

7.72 The electric power company is concerned with the loading of its transformers. Since it is responsible for a large number of customers, it must be certain that it can supply the demands of *all* customers. The power company's transformers will deliver rated kVA to the secondary load. However, if the demand increased to a point where greater than rated current were required, the secondary voltage would have to drop below rated value. Also, the current would increase, and with it the I^2R losses (due to winding resistance), possibly causing the transformer to overheat. Unreasonable current demand could be caused, for example, by excessively low power factors at the load.

The customer, on the other hand, is not greatly concerned with an inefficient power factor, provided that sufficient power reaches the load. To make the customer more aware of power factor considerations, the power company may install a penalty on the customer's bill. A typical penalty-power factor chart is shown in Table 7.3. Power factors below 0.7 are not permitted. A 25 percent penalty will be applied to any billing after two consecutive months in which the customer's power factor has remained below 0.7.

Table 7.3

Power factor	Penalty
0.850 and higher	None
0.8 to 0.849	1%
0.75 to 0.799	2%
0.7 to 0.749	3%

Courtesy of Detroit Edison.

The wye-wye circuit shown in Figure P7.72 is representative of a three-phase motor load.

- Find the total power supplied to the motor.
- Find the power converted to mechanical energy if the motor is 80 percent efficient.
- Find the power factor.
- Does the company risk facing a power factor penalty on its next bill if all the motors in the factory are similar to this one?

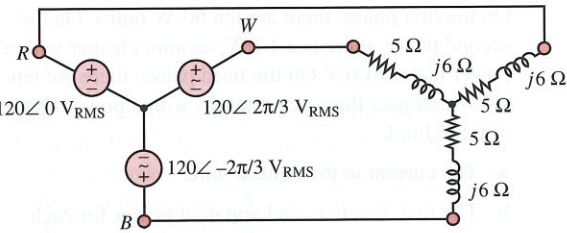


Figure P7.72

7.73 To correct the power factor problems of the motor in Problem 7.72, the company has decided to install capacitors as shown in Figure P7.73.

- What capacitance must be installed to achieve a unity power factor if the line frequency is 60 Hz?
- Repeat part a if the power factor is to be 0.85 lagging.

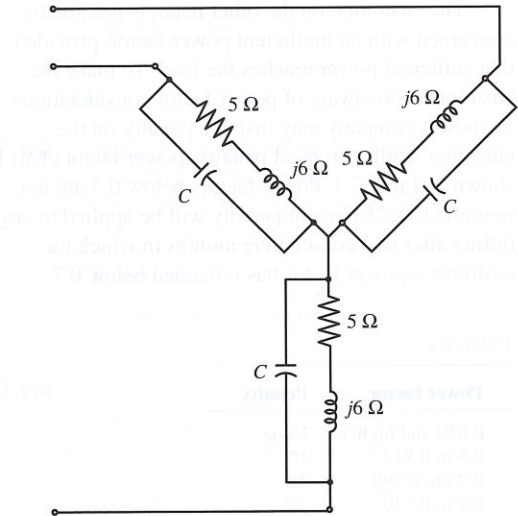


Figure P7.73

7.74 Find the apparent power and the real power delivered to the load in the Y-Δ circuit shown in Figure P7.74. What is the power factor?

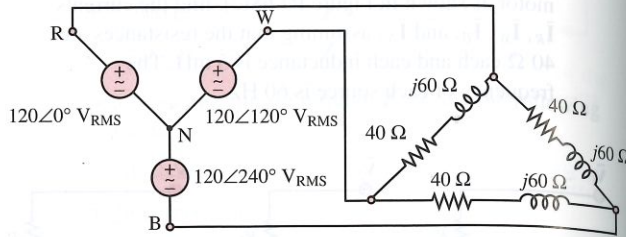


Figure P7.74

7.75 The circuit shown in Figure P7.75 is a Y-Δ-Y connected three-phase circuit. The primaries of the transformers are wye-connected, the secondaries are delta-connected, and the load is wye-connected. Find the currents $\tilde{I}_{RP}$, $\tilde{I}_{WP}$, $\tilde{I}_{BP}$, $\tilde{I}_A$, $\tilde{I}_B$, and $\tilde{I}_C$.

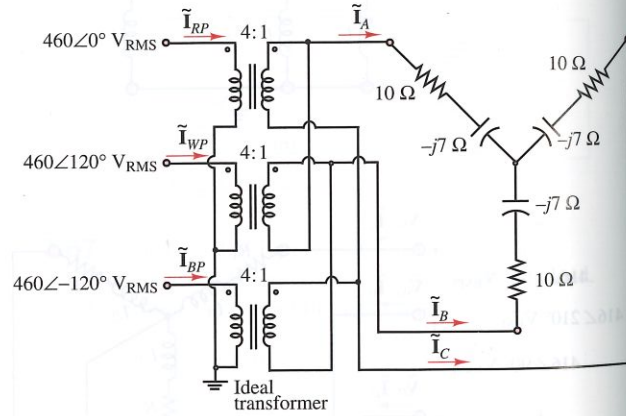


Figure P7.75

7.76 A three-phase motor is modeled by the wye-connected circuit shown in Figure P7.76. At $t = t_1$, a line fuse is blown (modeled by the switch). Find the line currents $\tilde{I}_R$, $\tilde{I}_W$, and $\tilde{I}_B$ and the power dissipated by the motor in the following conditions:

- $t \ll t_1$
- $t \gg t_1$

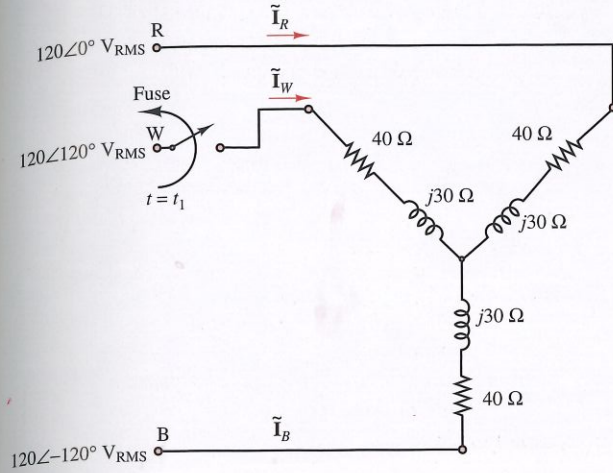


Figure P7.76

7.77 For the circuit shown in Figure P7.77, find the currents $\tilde{I}_A$, $\tilde{I}_B$, $\tilde{I}_C$ and $\tilde{I}_N$, and the real power dissipated by the load.

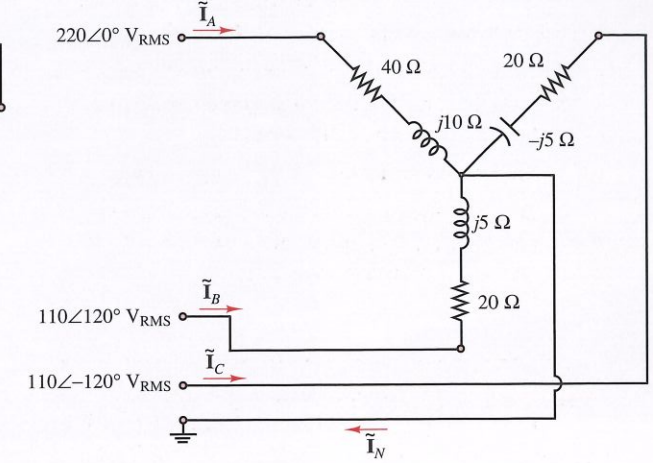


Figure P7.77